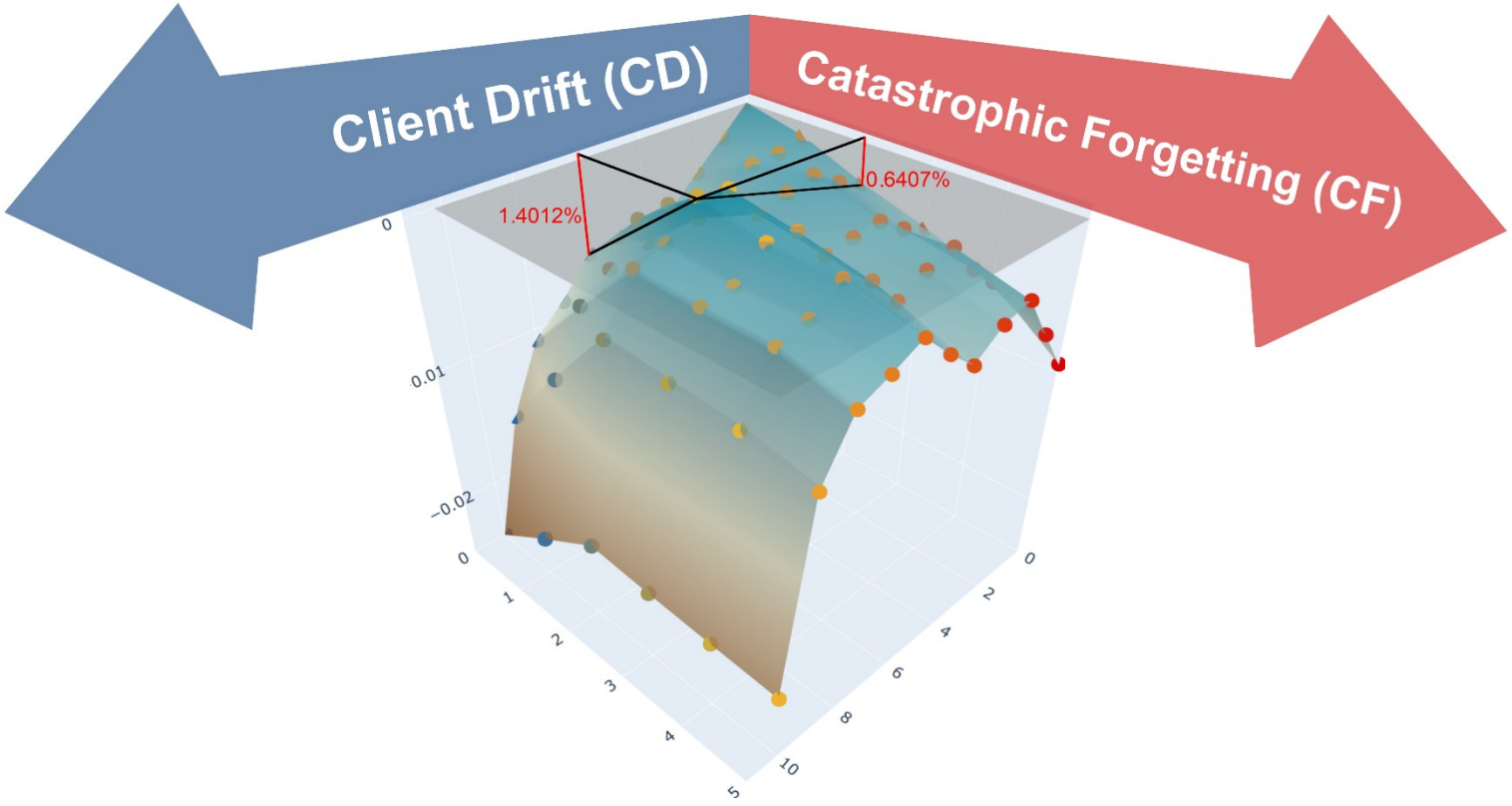
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Dynamic Learning
in space-time continuum

Anirban Mukhopadhyay

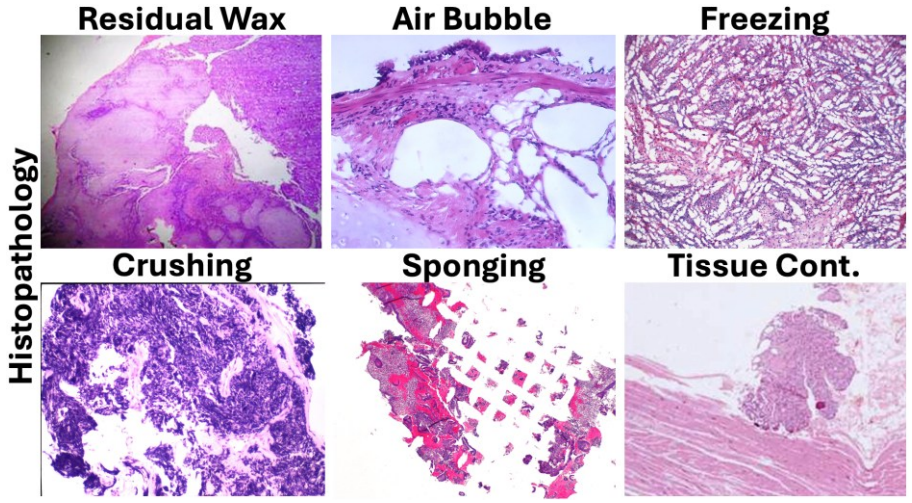


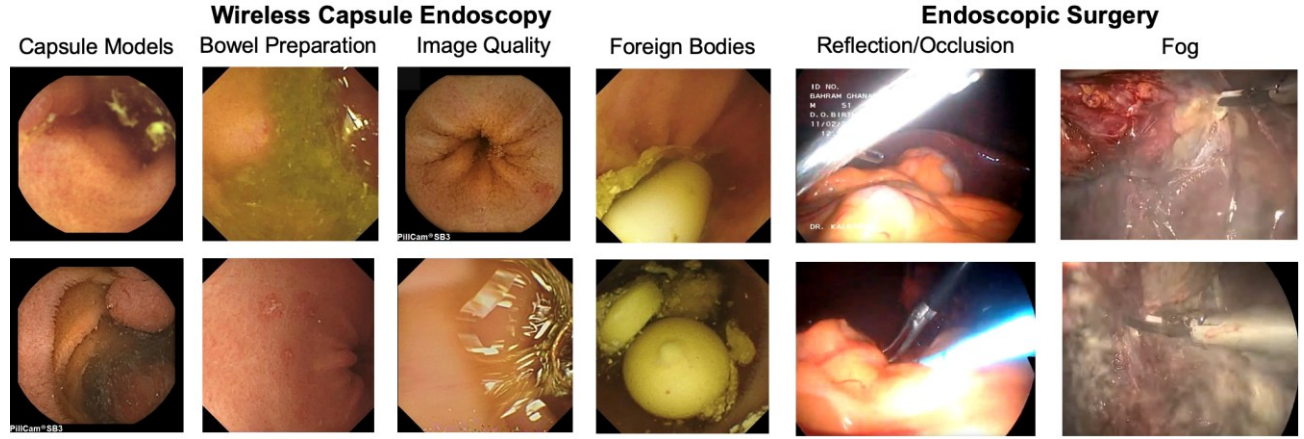
Motivation

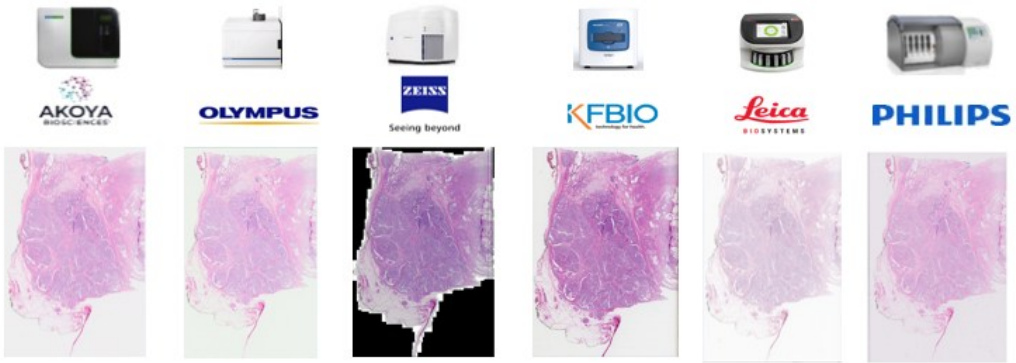
The Dynamic Clinical Reality of Medical Imaging

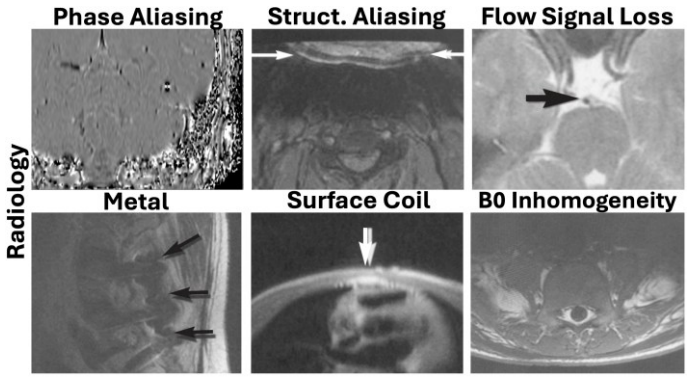
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The Dynamic Reality of Medical Imaging







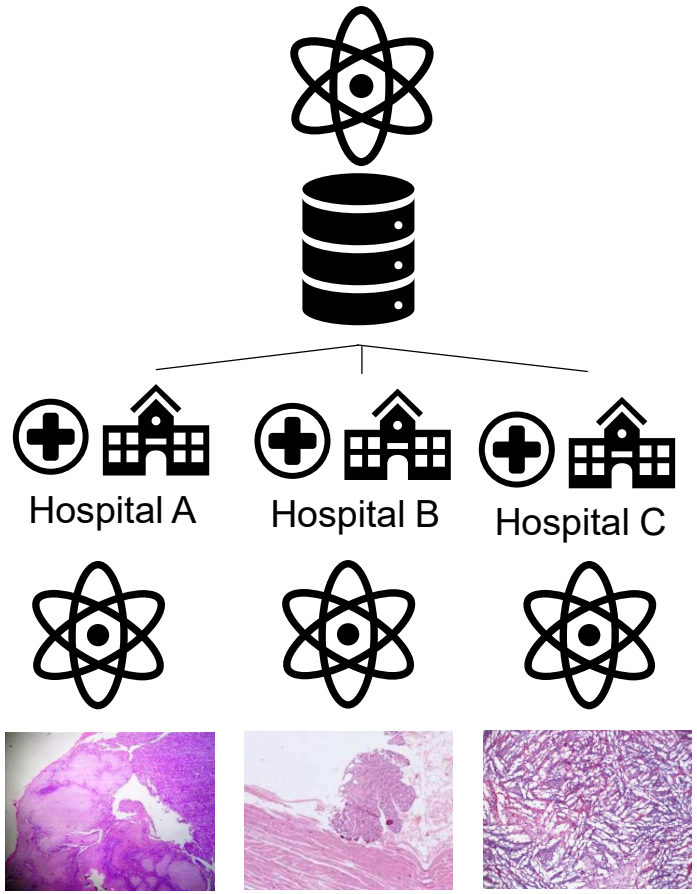


and many more…

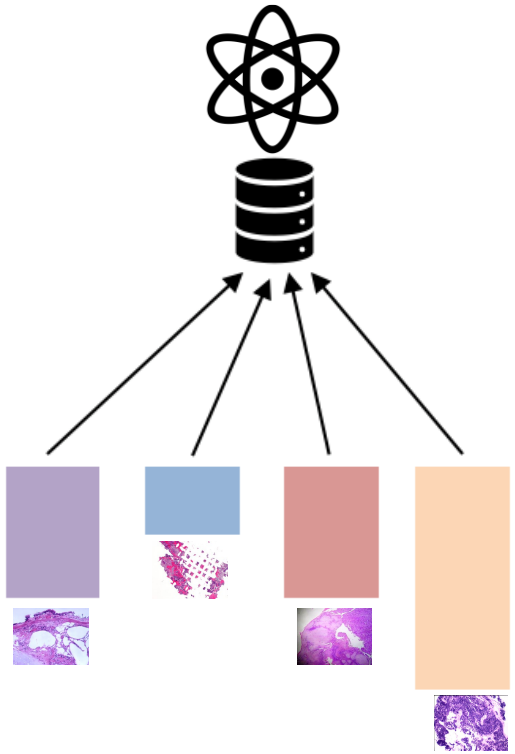
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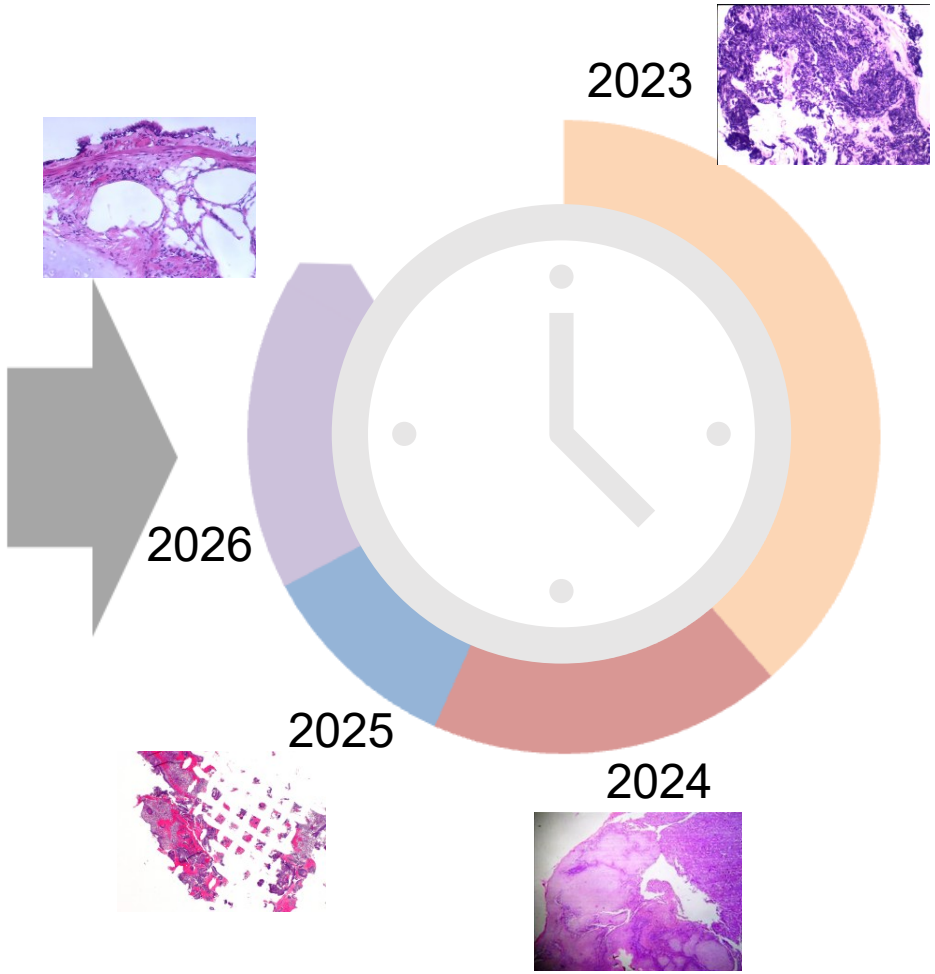
Solutions for Data (Un-)availability

Space: Federated Learning



Time: Continual Learning





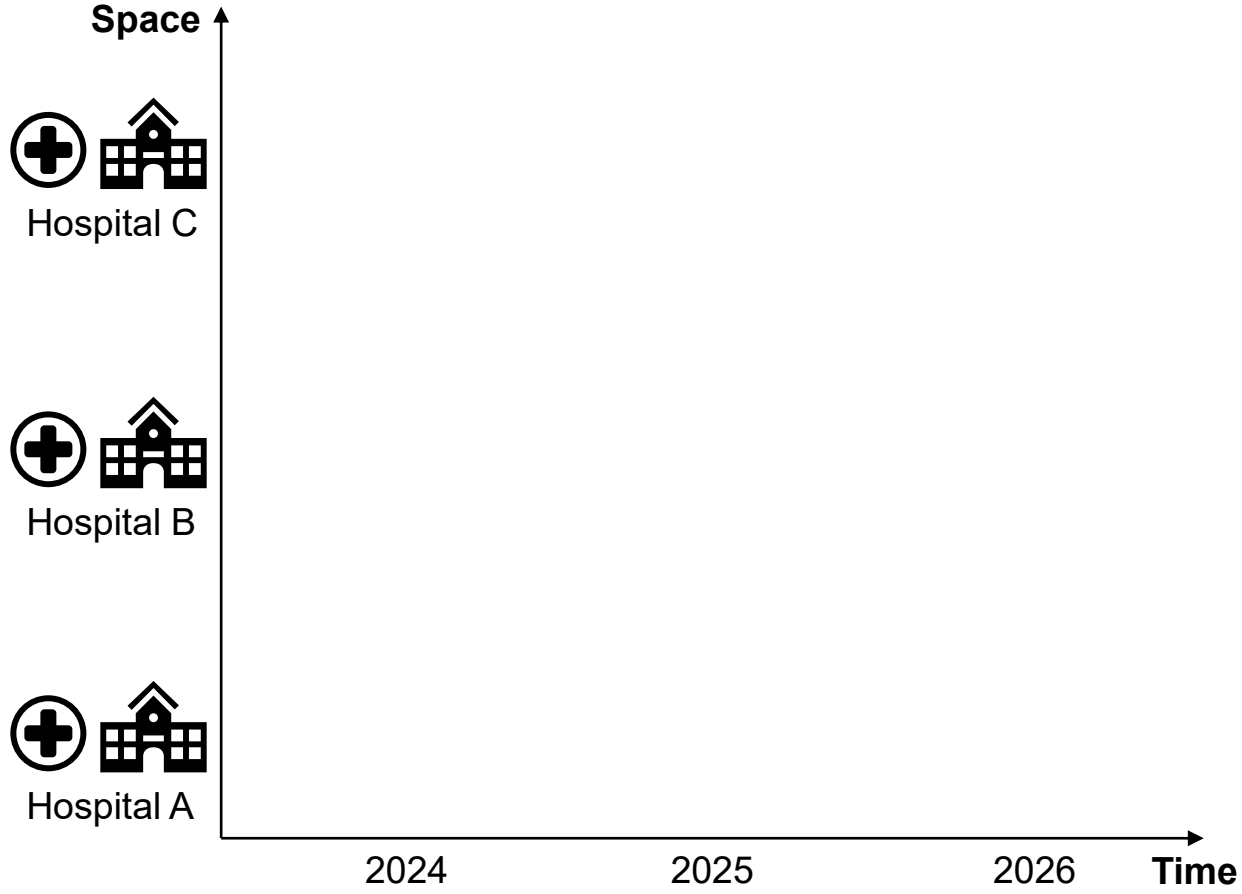
The Data Availability Gap

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Naive Approach: Centralized Training



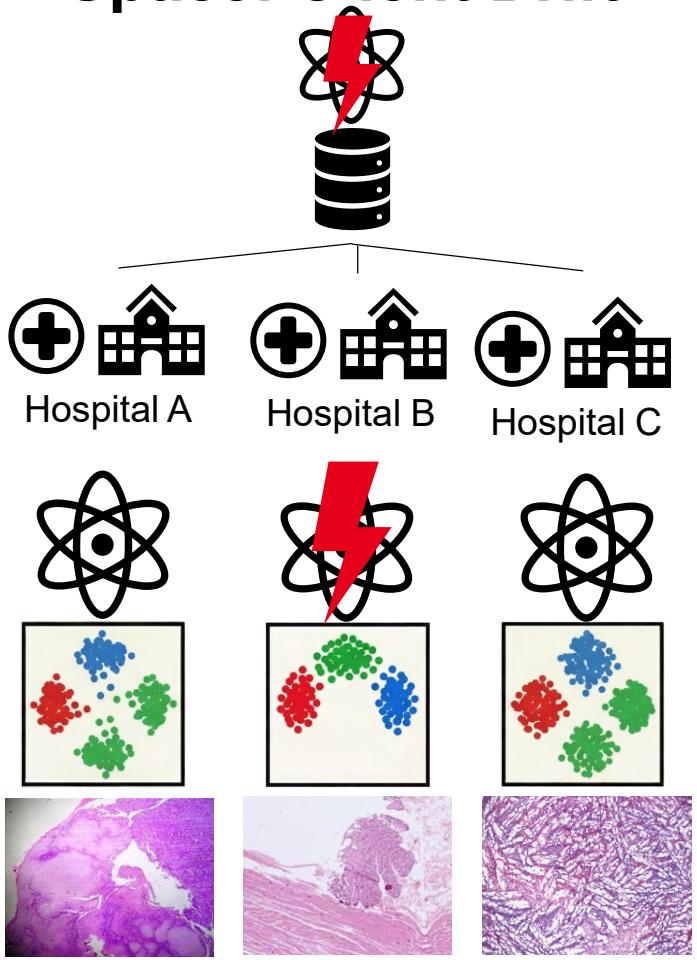
Distributed Reality:
Spatio-temporally distributed due to privacy regulations (GDPR/HIPAA)



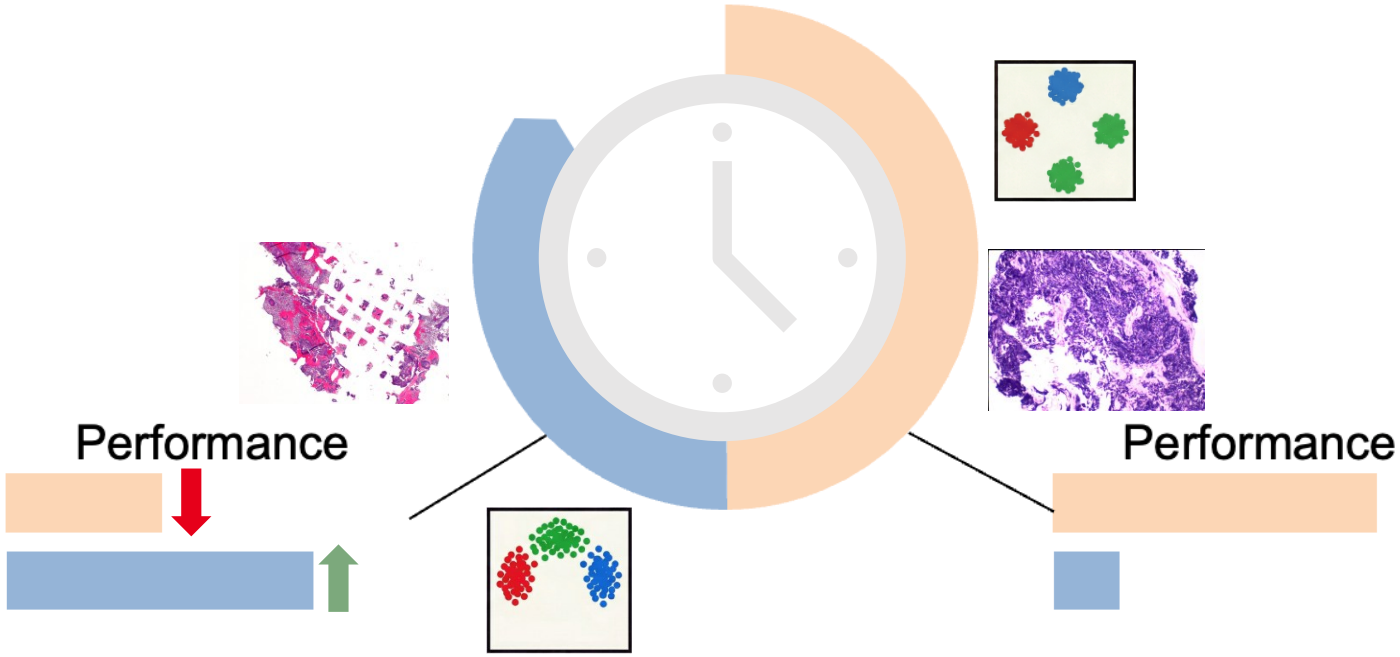
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Issues with Distributed Training

Space: Client Drift



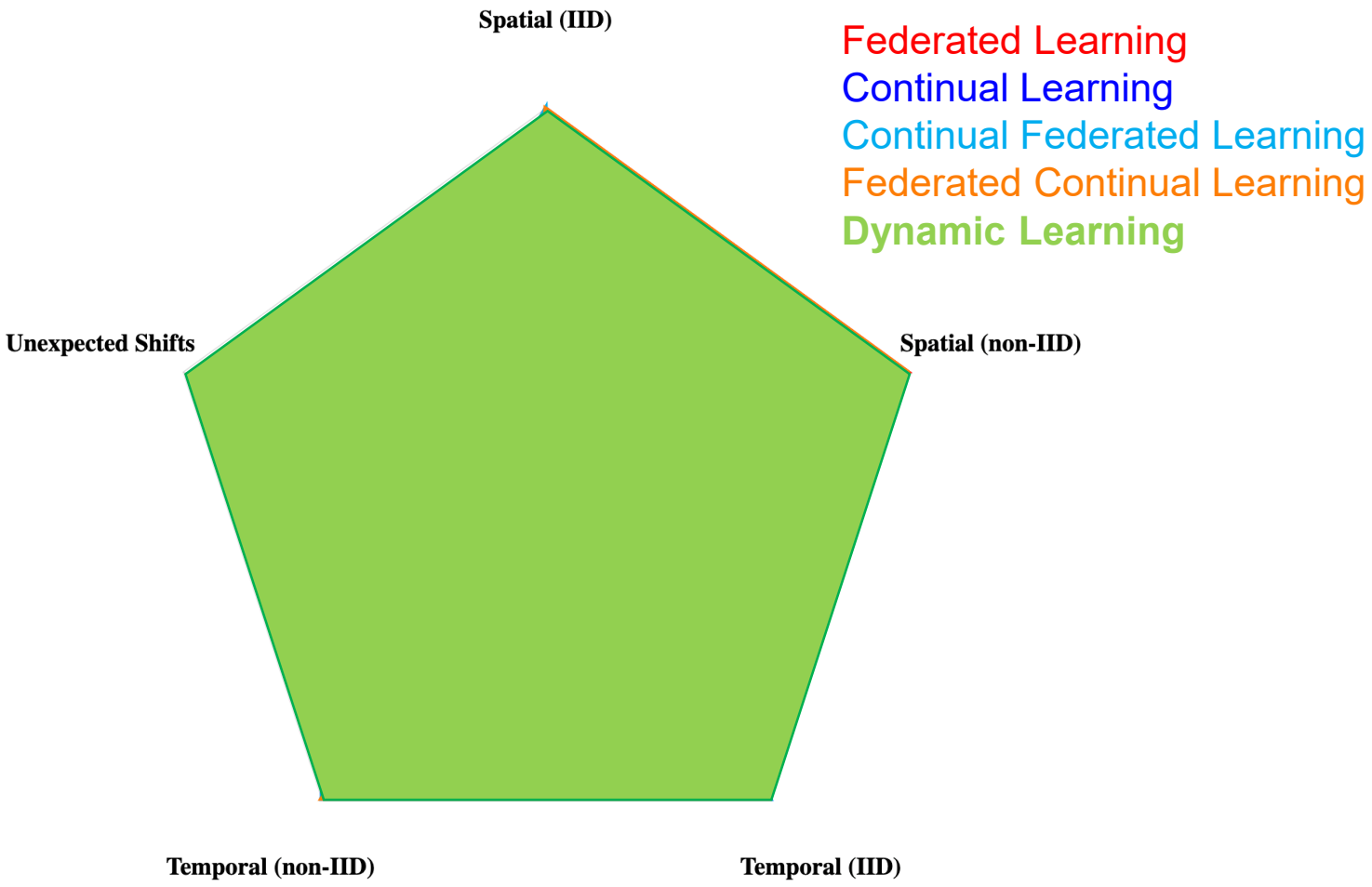
Time: Catastrophic Forgetting



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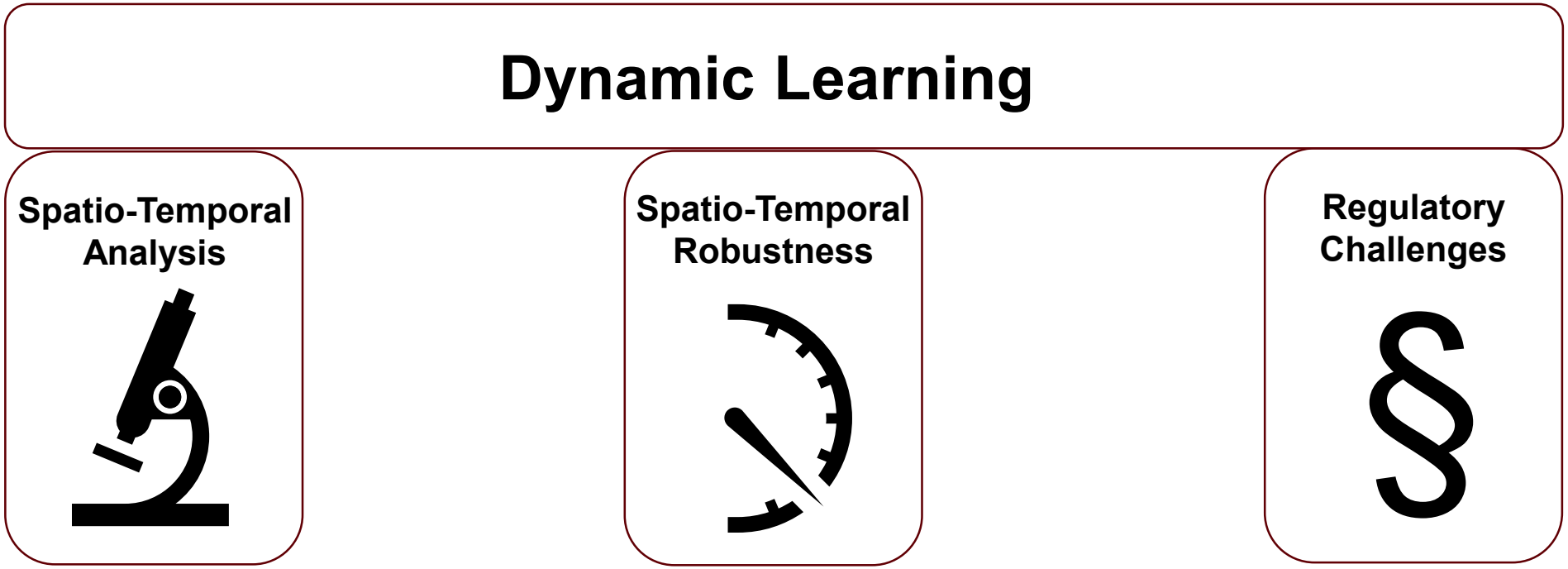
Spatio-Temporal Gap Persists

Learning Paradigm	Solves Spatial Shifts (CD)	Solves Temporal Shifts (CF)	Task-Agnostic (Unexpected Shift)	Approach	Method	Year
Federated Learning	✓	×	(✓)	Baseline	*FedAvg* [63]	2017
				Outliers	*FedAvgM* [42]	2019
					FedProx [53]	2020
				Compensate	*SCAFFOLD* [46]	2020
					AdaBest [94]	2022
					Xu et al. [97]	2025
				Personaliz.	*FedBABU* [68]	2021
				Knowledge Distill.	*FedDF* [55]	2020
					FedMD [51]	2019
					FedDyn [45]	2023
				Prototype	*FedProto* [87]	2022
				Contrastive	*FedPCL* [86]	2022
					FedProc [66]	2023
				Param. Tuning	*FedPop* [17]	2025
				Data Sharing	*FedMatch* [44]	2021



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Structure



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Dynamic Learning

Spatio-Temporal Analysis

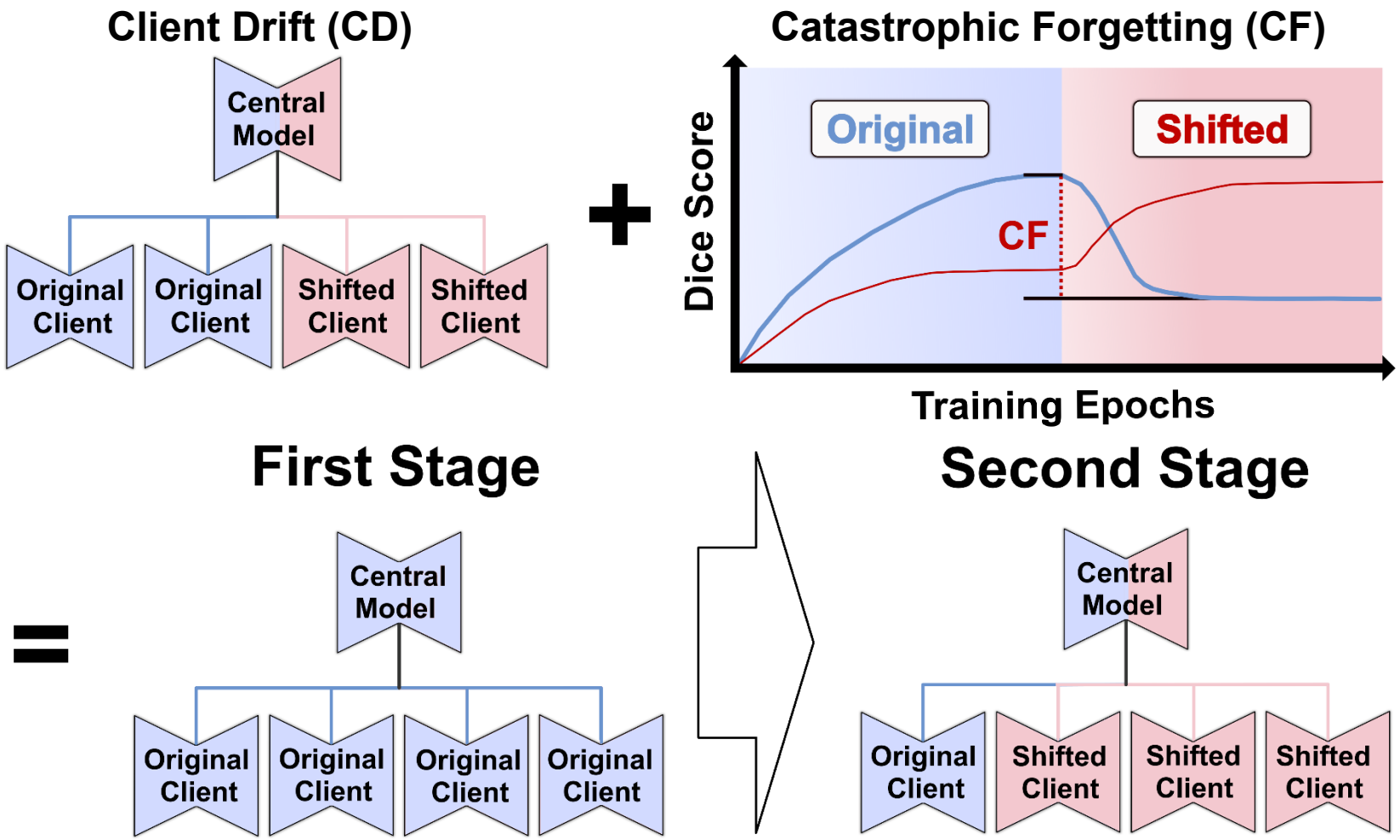


Spatio-Temporal Analysis

Correlation and Interaction of Client Drift and Catastrophic Forgetting

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Spatio-Temporal Analysis



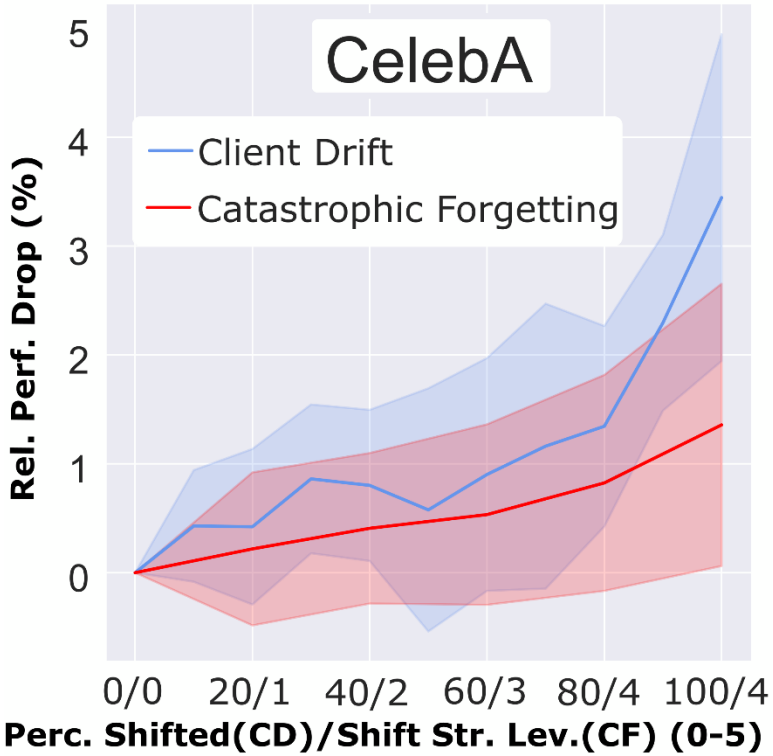


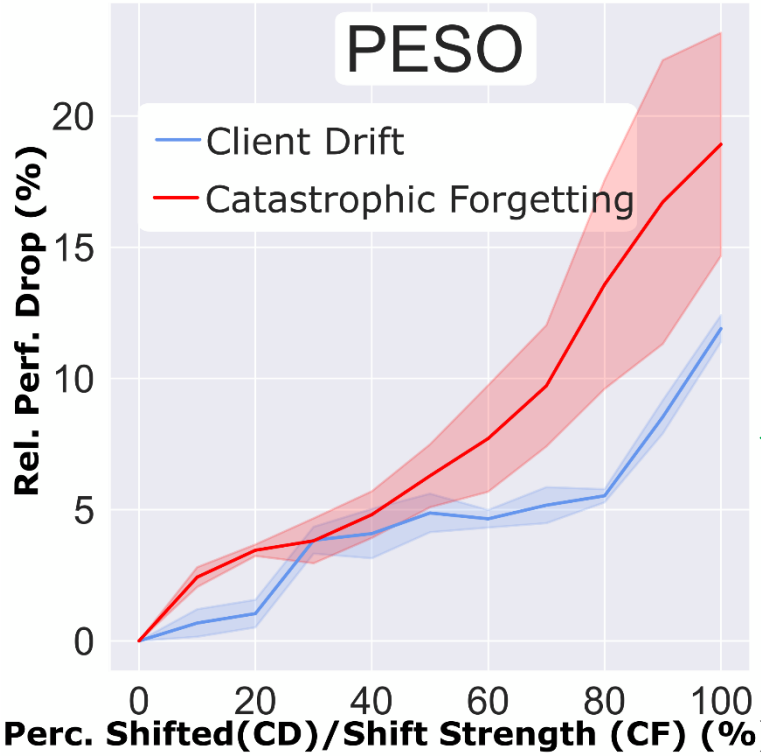
Babendererde, N., Fuchs, M., Gonzalez, C., Tolkach, Y., & Mukhopadhyay, A. (2025). Jointly exploring client drift and catastrophic forgetting in dynamic learning. *Scientific Reports*, *15*(1), 5857.

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Spatio-Temporal Analysis: Correlation

Hypothesis 1 (H1) *The performance drop through CD Δp_{CD}, caused by a certain share of shifted clients $\frac{s_{\hat{\alpha}}}{c}$ with a shift strength $\hat{\alpha}$, is correlated to the drop from Catastrophic Forgetting Δp_{CF} that is caused by a certain shift strength α.* $\exists m : \Delta p_{CD} = f(\frac{s_{\hat{\alpha}}}{c}) \sim m \cdot g(\alpha) = m$





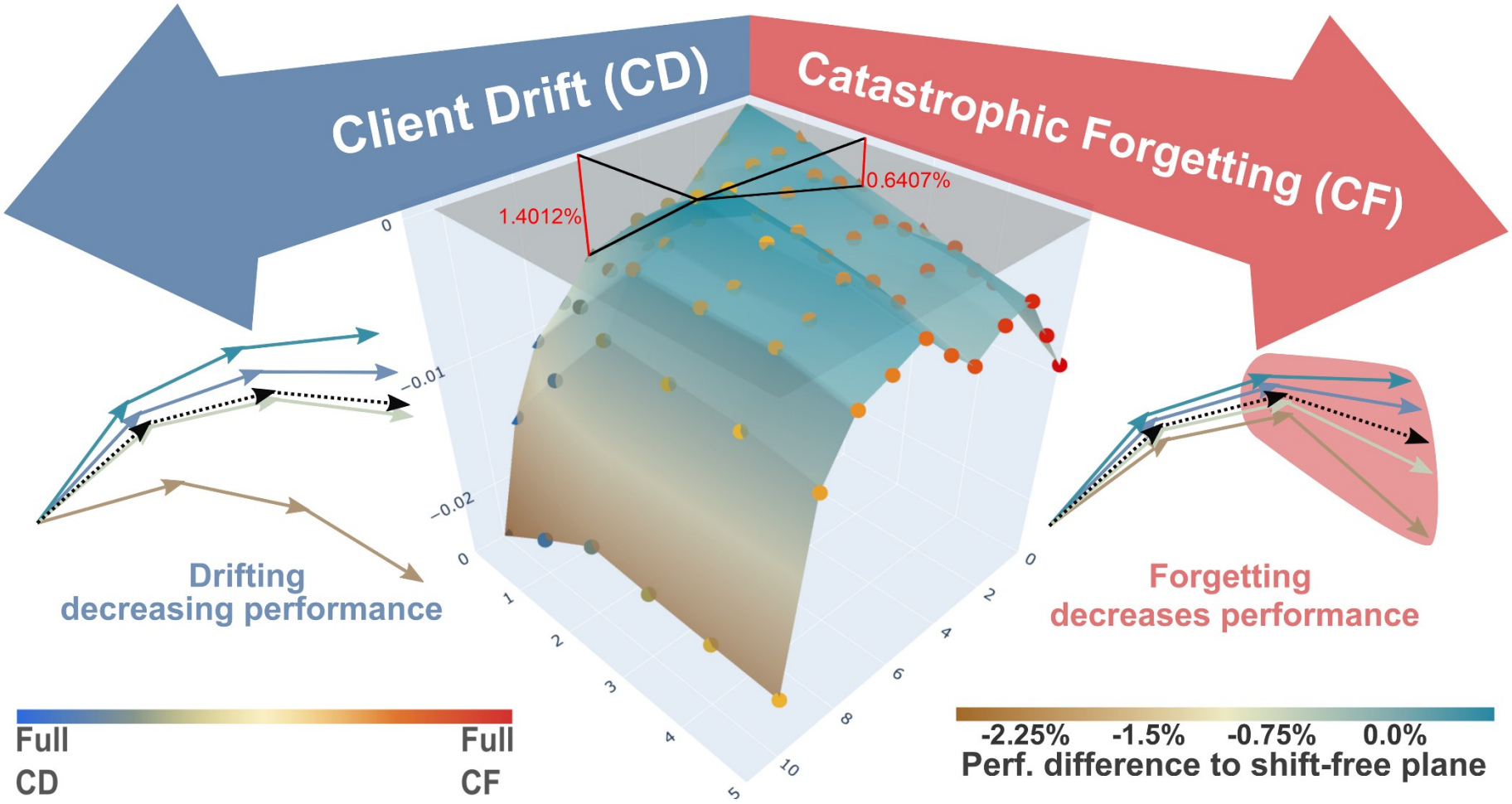
Spearman's Rank CE:
CelebA: 1.0
PESO: 0.9909

✓ Piece-wise linear correlation

Babendererde, N., Fuchs, M., Gonzalez, C., Tolkach, Y., & Mukhopadhyay, A. (2025). Jointly exploring client drift and catastrophic forgetting in dynamic learning. *Scientific Reports*, *15*(1), 5857.

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Spatio-Temporal Analysis: Interaction between CD and CF

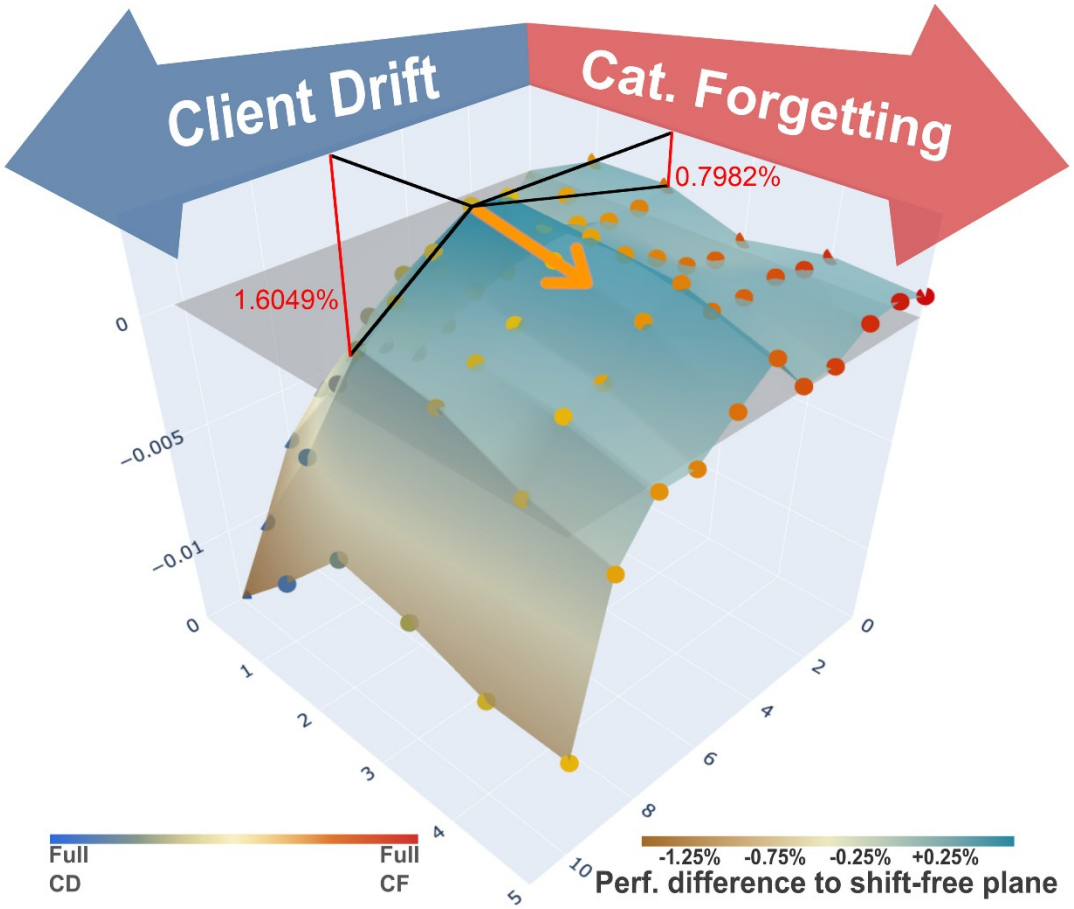


Babendererde, N., Fuchs, M., Gonzalez, C., Tolkach, Y., & Mukhopadhyay, A. (2025). Jointly exploring client drift and catastrophic forgetting in dynamic learning. *Scientific Reports*, *15*(1), 5857.

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Spatio-Temporal Analysis: Interaction between Dimensions

Rehearsal:
Method against CF that saves a certain number of training samples between epochs partially trains on them besides the current training data



Babendererde, N., Fuchs, M., Gonzalez, C., Tolkach, Y., & Mukhopadhyay, A. (2025). Jointly exploring client drift and catastrophic forgetting in dynamic learning. *Scientific Reports*, *15*(1), 5857.

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Dynamic Learning

Spatio-Temporal Analysis



Spatio-Temporal Robustness

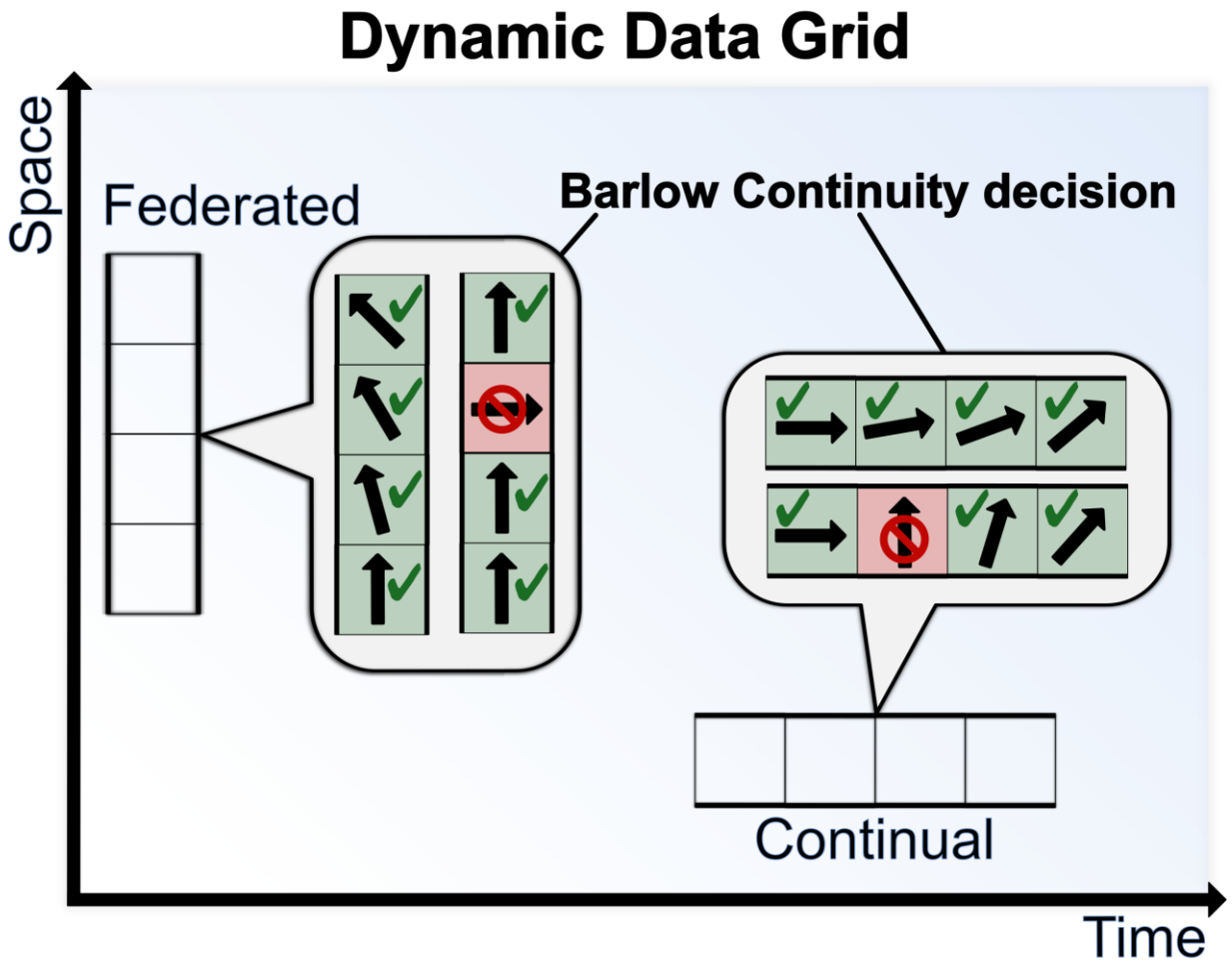


Joint Method for ST-Robustness

Introducing a Joint Method for Dynamic Learning

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Joint Method for ST-Robustness: Intuition



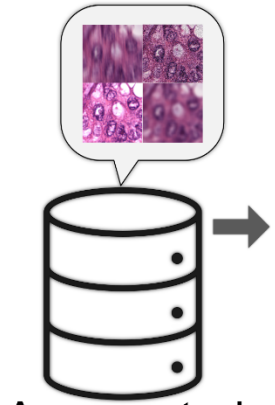


Babendererde, N., Zhu, H., Fuchs, M., Stieber, J., & Mukhopadhyay, A. (2025, February). Federated-continual dynamic segmentation of histopathology guided by barlow continuity. In *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)* (pp. 3752-3761). IEEE.

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Joint Method for ST-Robustness: Method

1. Dynamic Barlow Continuity (DynBC)



$$DynBC = \frac{\sum_{n=0}^{N} \langle m_s(a(x_n)), m_c(a(x_n)) \rangle}{N}$$

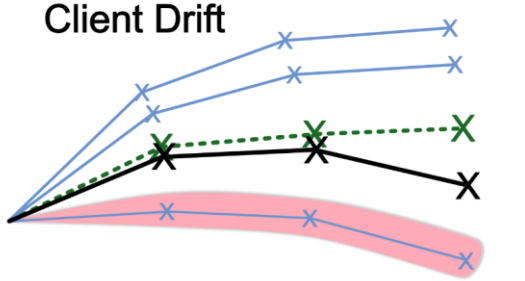
Babendererde, N., Zhu, H., Fuchs, M., Stieber, J., & Mukhopadhyay, A. (2025, February). Federated-continual dynamic segmentation of histopathology guided by barlow continuity. In *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)* (pp. 3752-3761). IEEE.

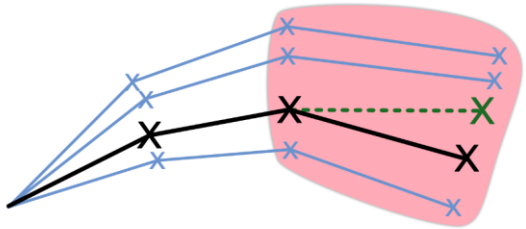
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Joint Method for ST-Robustness: Intuition

DynBC reduces CD/CF

Federated Learning:
Client Drift



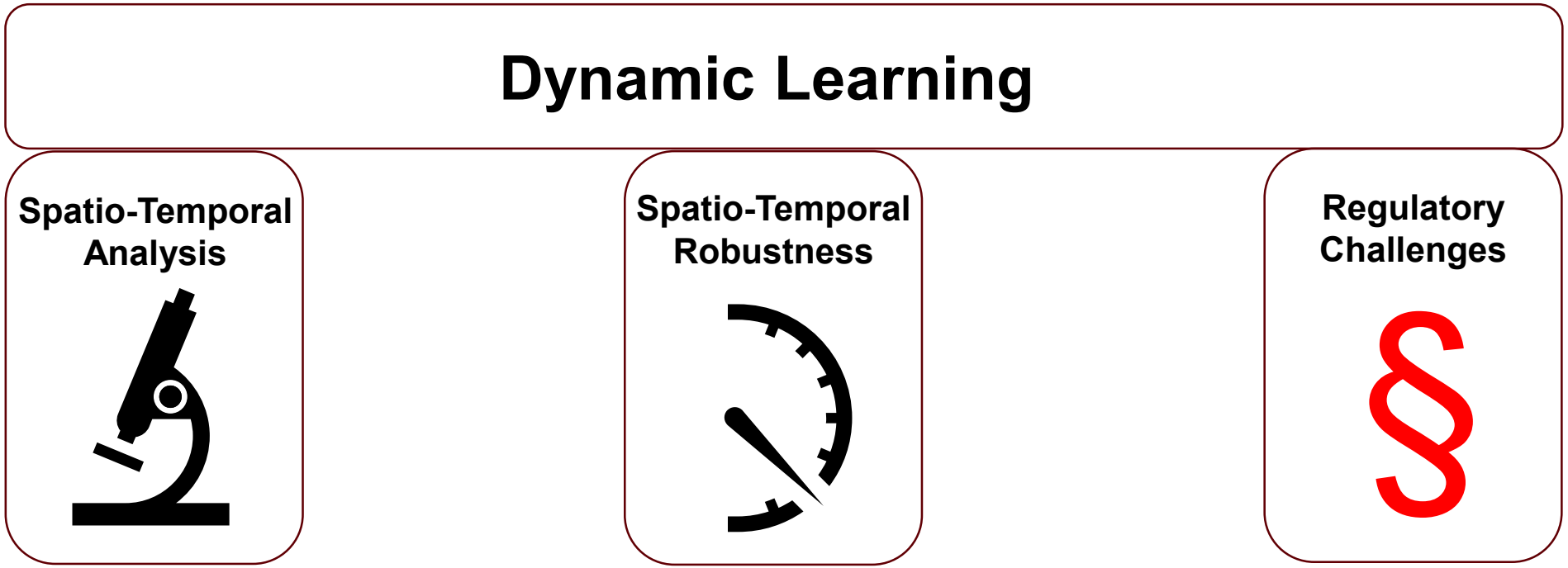


Continual Learning:
Catastrophic Forgetting

Dataset	BCSS (Dice Scores)		Semicol (Dice Scores)	
Shift Type	Gaussian Blur	Brightness	Gaussian Blur	Brightness
FedAvg (FL, Baseline)	0.579 ± 0.062	0.479 ± 0.151	0.077 ± 0.058	0.2 ± 0.129
DynBC (Ours)	$\mathbf{0.698 \pm 0.21}$	$\mathbf{0.731 \pm 0.118}$	$\mathbf{0.518 \pm 0.226}$	$\mathbf{0.733 \pm 0.031}$
Rehearsal (CL)	0.669 ± 0.106	0.682 ± 0.1	0.353 ± 0.231	0.328 ± 0.212
FedAdam (FL)	0.553 ± 0.479	0.553 ± 0.479	0.014 ± 0.024	0.008 ± 0.012

Babendererde, N., Zhu, H., Fuchs, M., Stieber, J., & Mukhopadhyay, A. (2025, February). Federated-continual dynamic segmentation of histopathology guided by barlow continuity. In *2025 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)* (pp. 3752-3761). IEEE.

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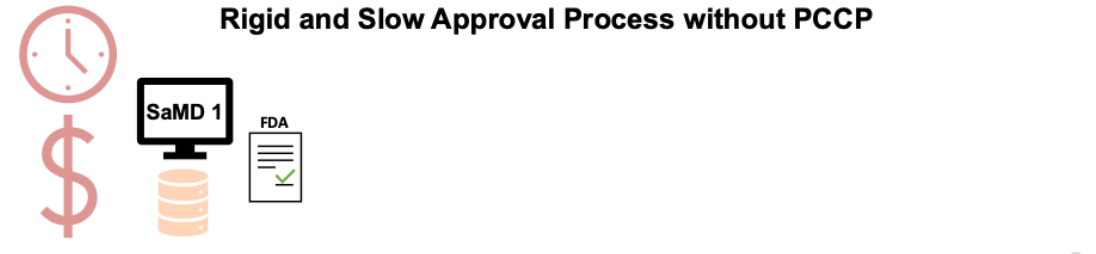


Regulatory Challenges for Dynamic Learning

Analyzing FDA‘s Predetermined Change Control Plan regarding Opportunities and Gaps in the Context of Dynamic Learning

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Regulating Dynamic Learning



Retraining without new device registration

Simplified re-approval process

Examples

?

Evaluation Gap

Intended Use Gap

Foundation Model Gap

Mandate to evaluate on **previous distributions**

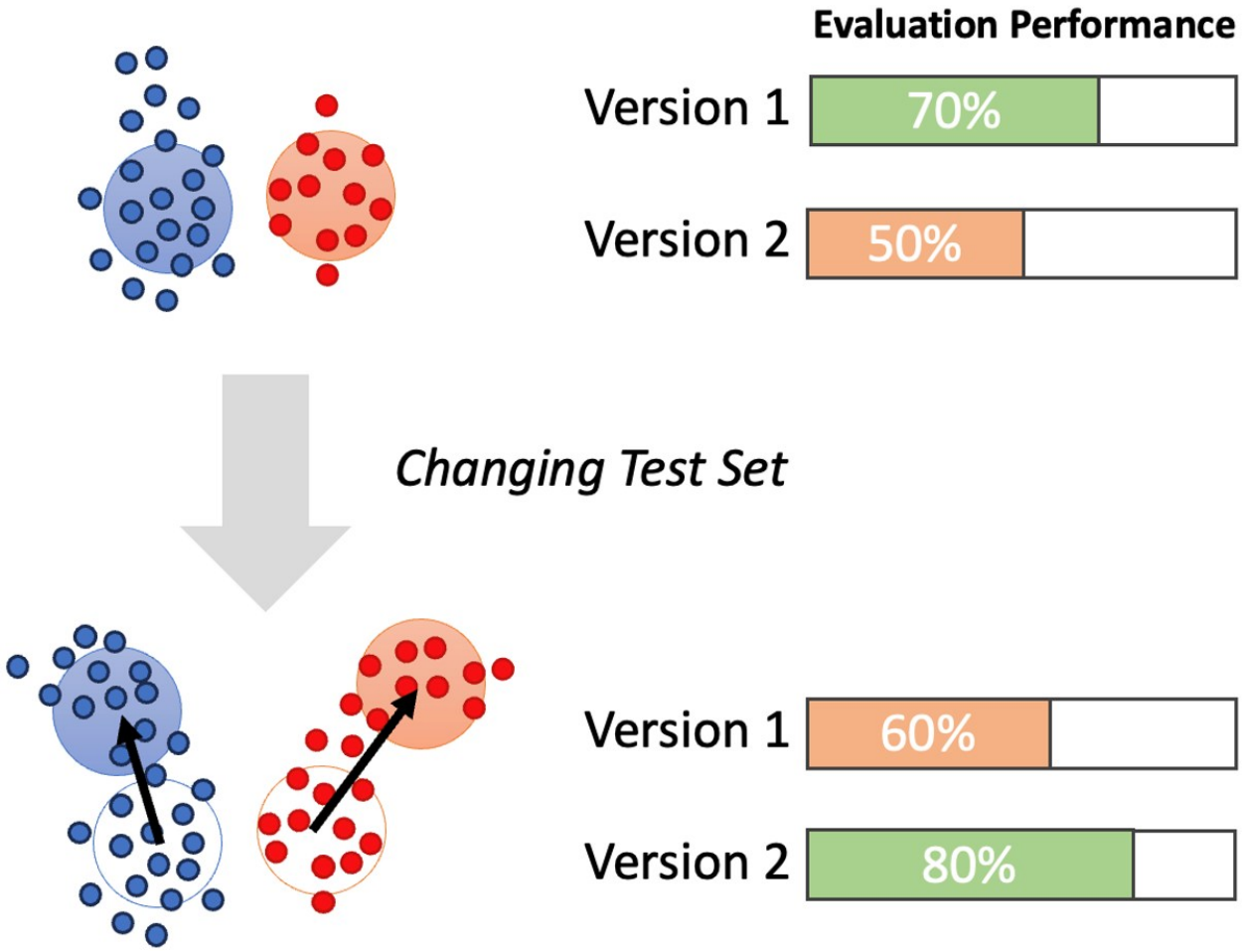
Allowing new diseases with **shared manifestations**

Allow **groups of tasks and modalities**

Babendererde, N., Ranem, A., Fuchs, M., González, C., Krumb, H. J., & Mukhopadhyay, A. (2025, September). FDA's PCCP: Opportunities and Gaps. In *International Conference on Medical Image Computing and Computer-Assisted Intervention* (pp. 56-64). Cham: Springer Nature Switzerland.

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Evaluation Gap



Babendererde, N., Ranem, A., Fuchs, M., González, C., Krumb, H. J., & Mukhopadhyay, A. (2025, September). FDA's PCCP: Opportunities and Gaps. In *International Conference on Medical Image Computing and Computer-Assisted Intervention* (pp. 56-64). Cham: Springer Nature Switzerland.

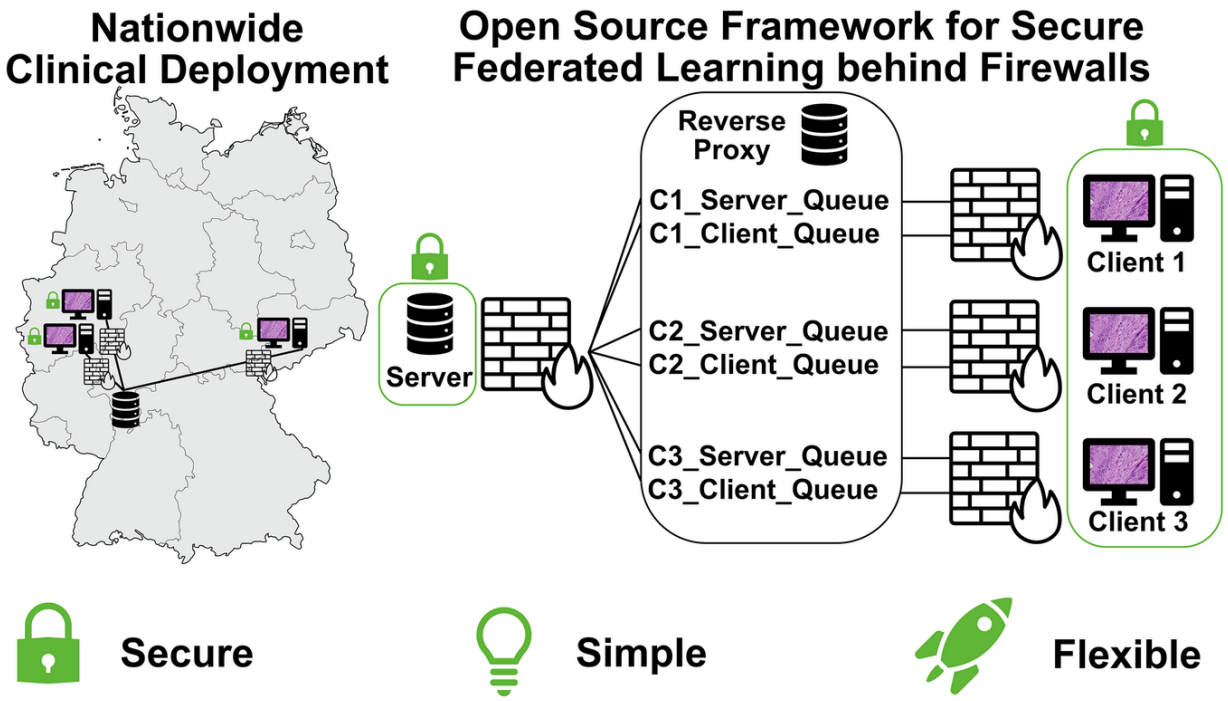
Solution:

- Define minimum performance in marketing submission
- Each update must maintain this minimum performance additionally on all previous test set versions



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Open-Source Software

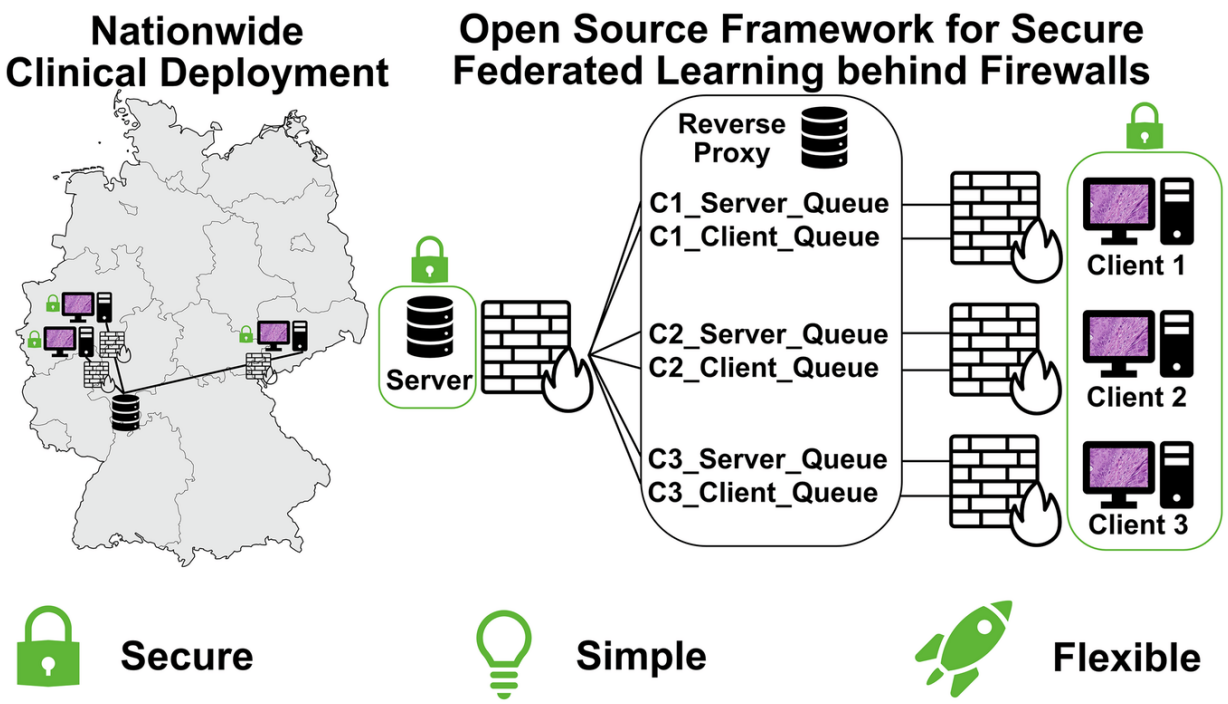


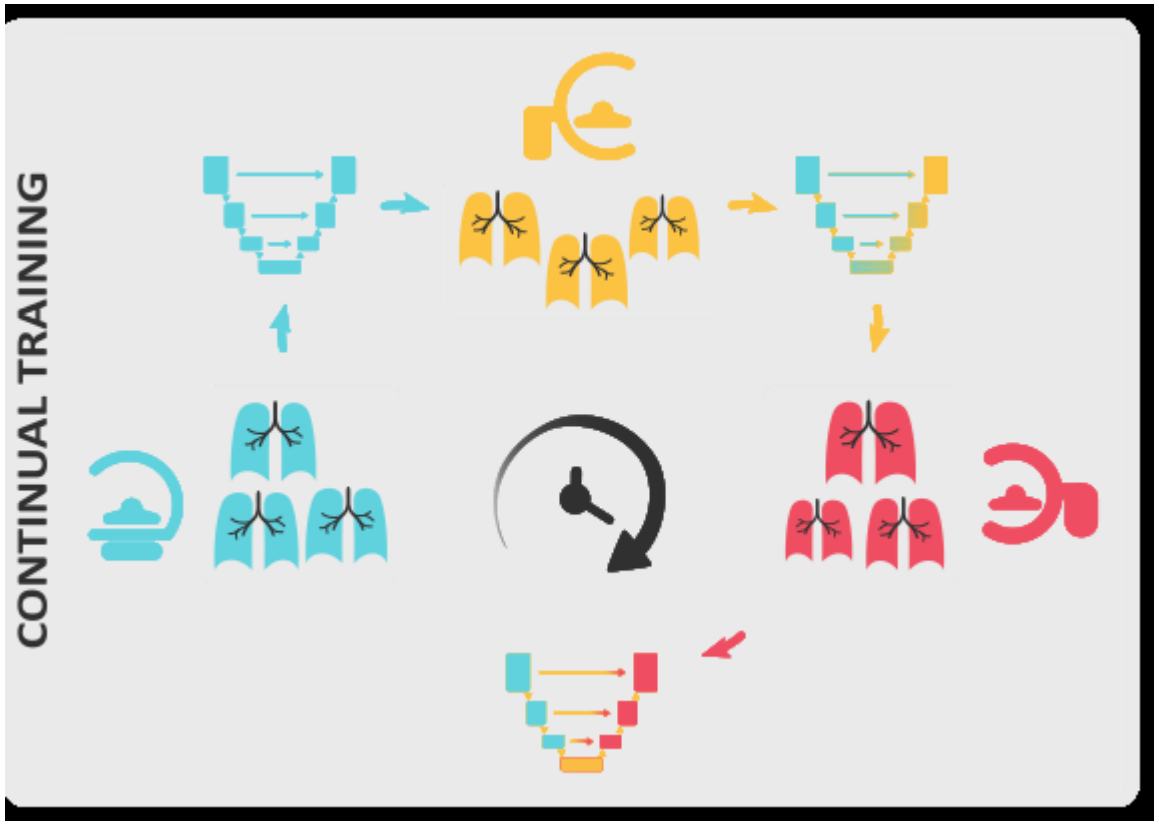
https://www.nature.com/articles/s41746-026-02958-y

https://github.com/MECLabTUDA/TheODen

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Open-Source Software





https://www.nature.com/articles/s41746-026-02958-y

https://github.com/MECLabTUDA/TheODen

https://github.com/MECLabTUDA/Lifelong-nnUNet

SUMMARY

Data shift and (Un-)availability as the first category problem of Medical AI

Dynamic Learning

Spatio-Temporal Analysis



Spatio-Temporal Robustness



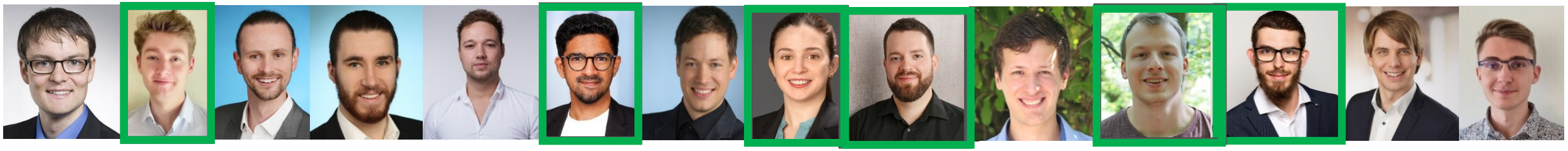
Regulatory Challenges



- ✓ Client Drift and Catastrophic Forgetting strongly correlated
- ✓ Introduced **Dynamic Learning** and a joint method mitigating both
- ✓ Regulatory Challenges of Dynamic Learning

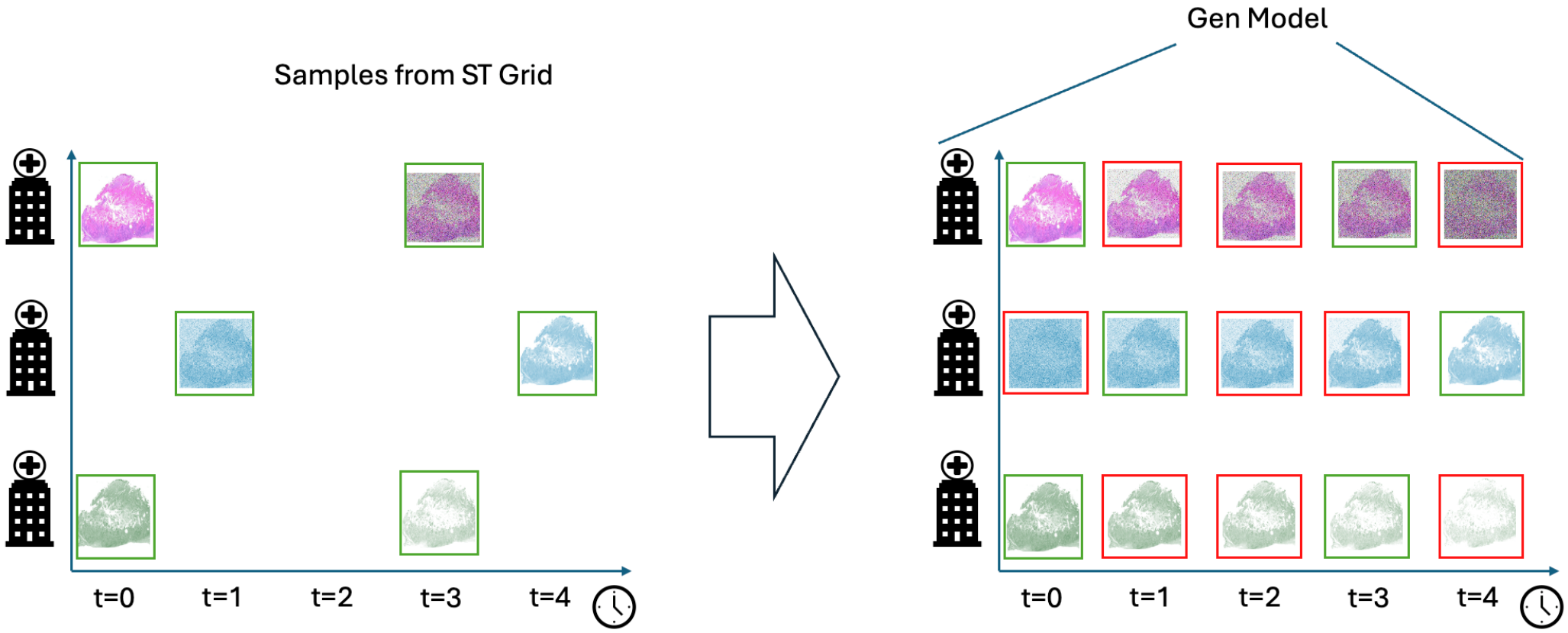


LinkedIn



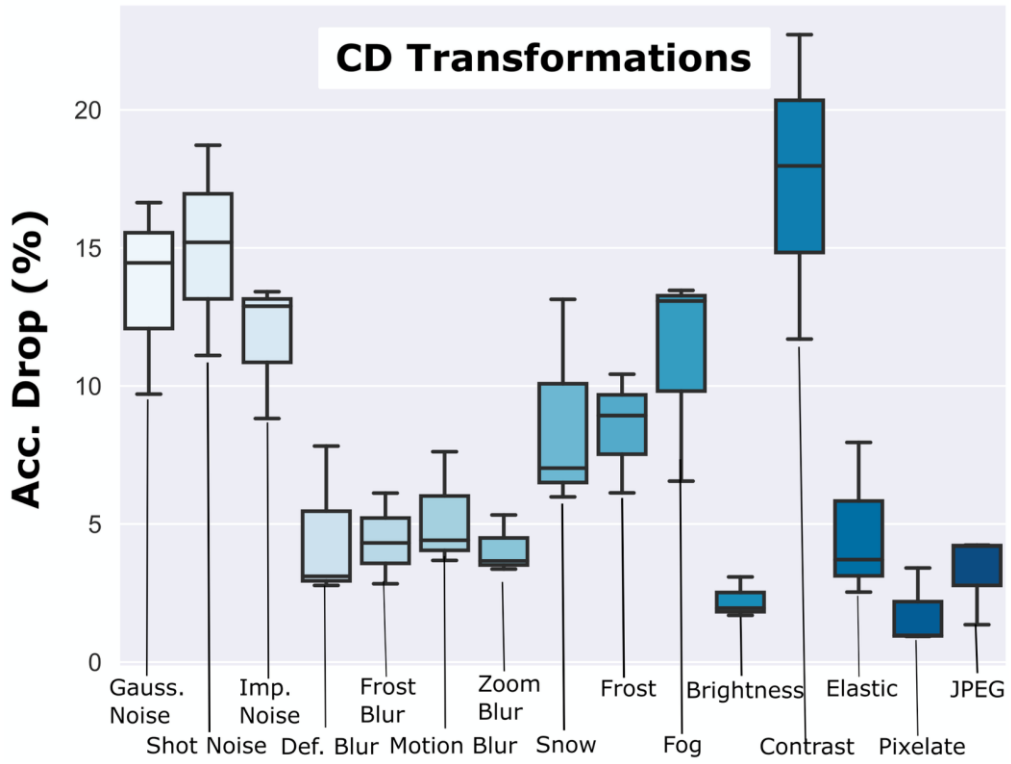
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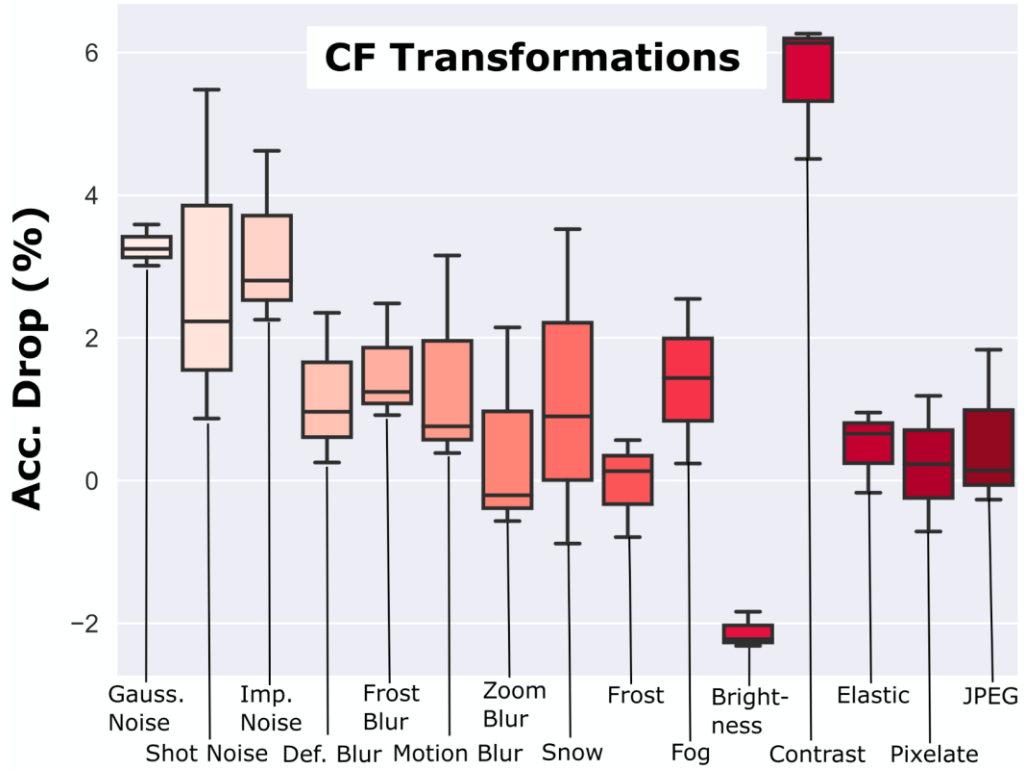
Outlook



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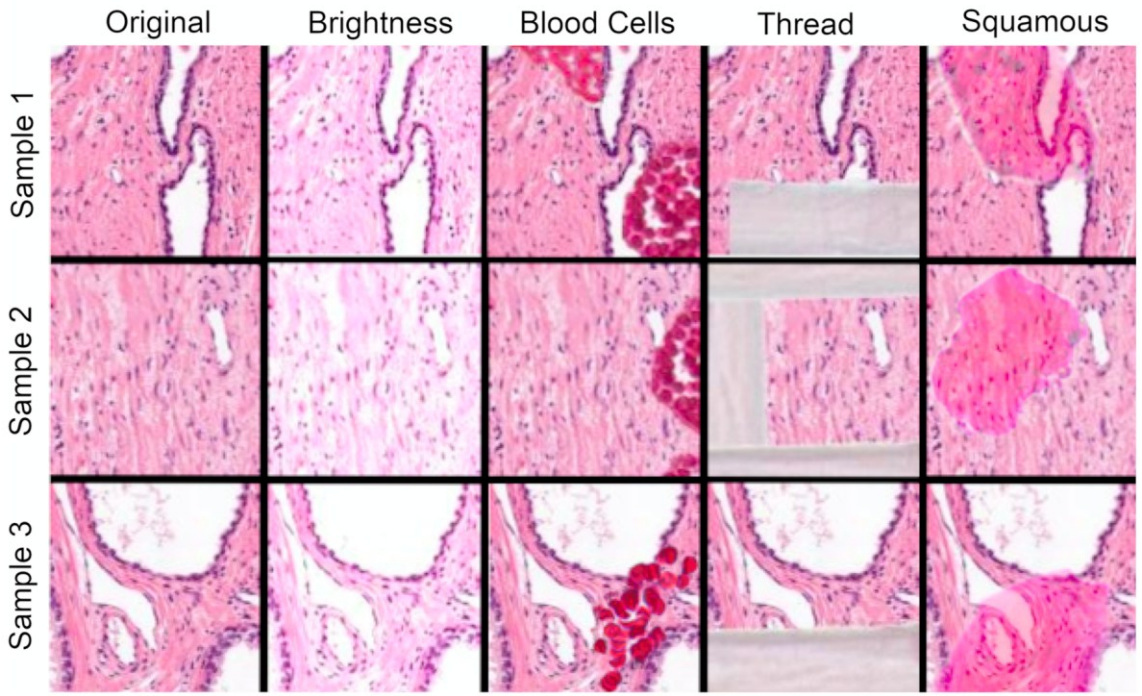
ST-Analysis: Augmentations

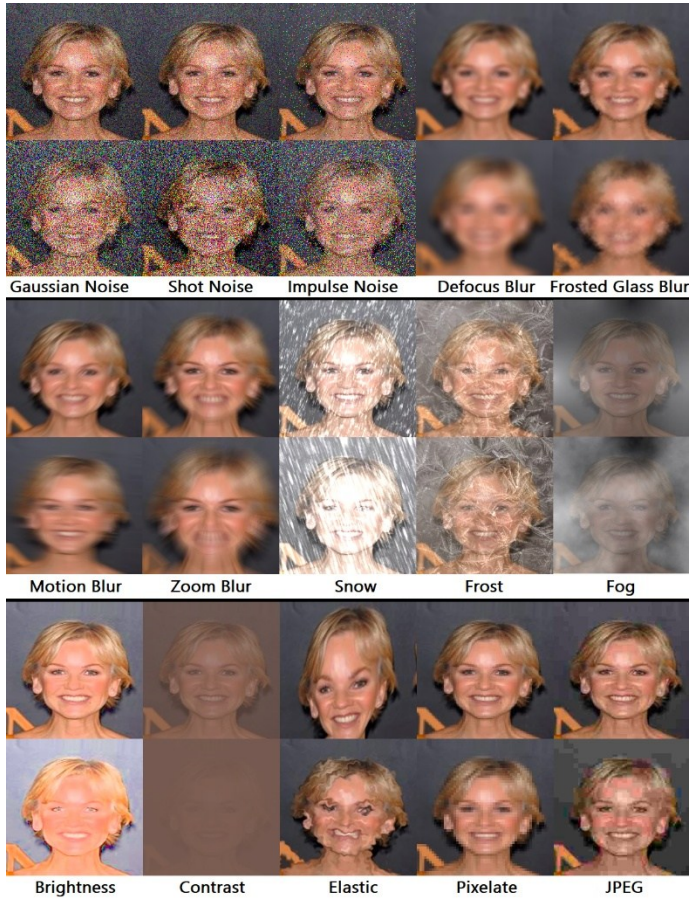




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ST-Analysis: Augmentations



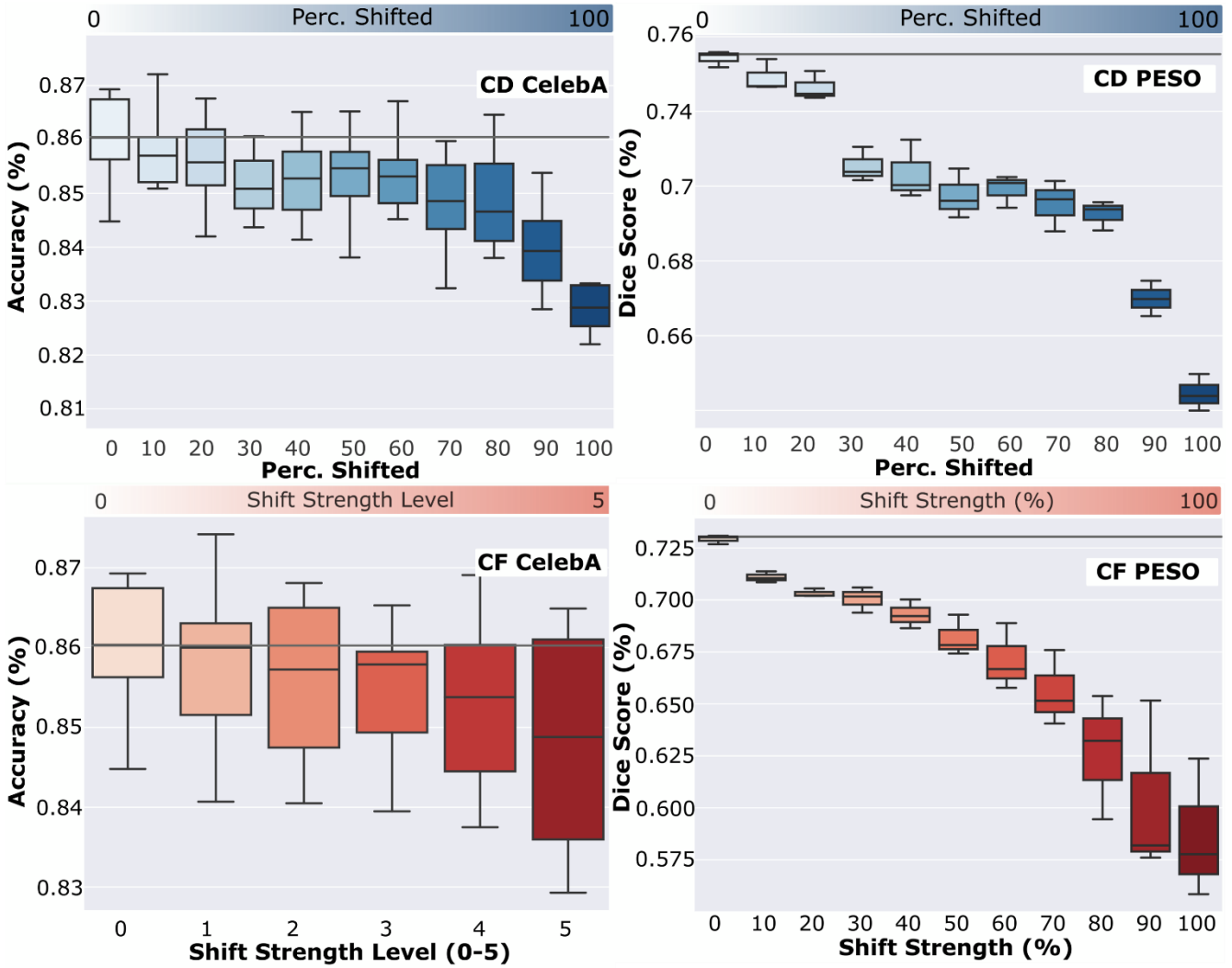


ST-Analysis: Correlation

	CelebA	PESO
Spearman	1.0	0.9909
Pearson	0.9712	0.9385

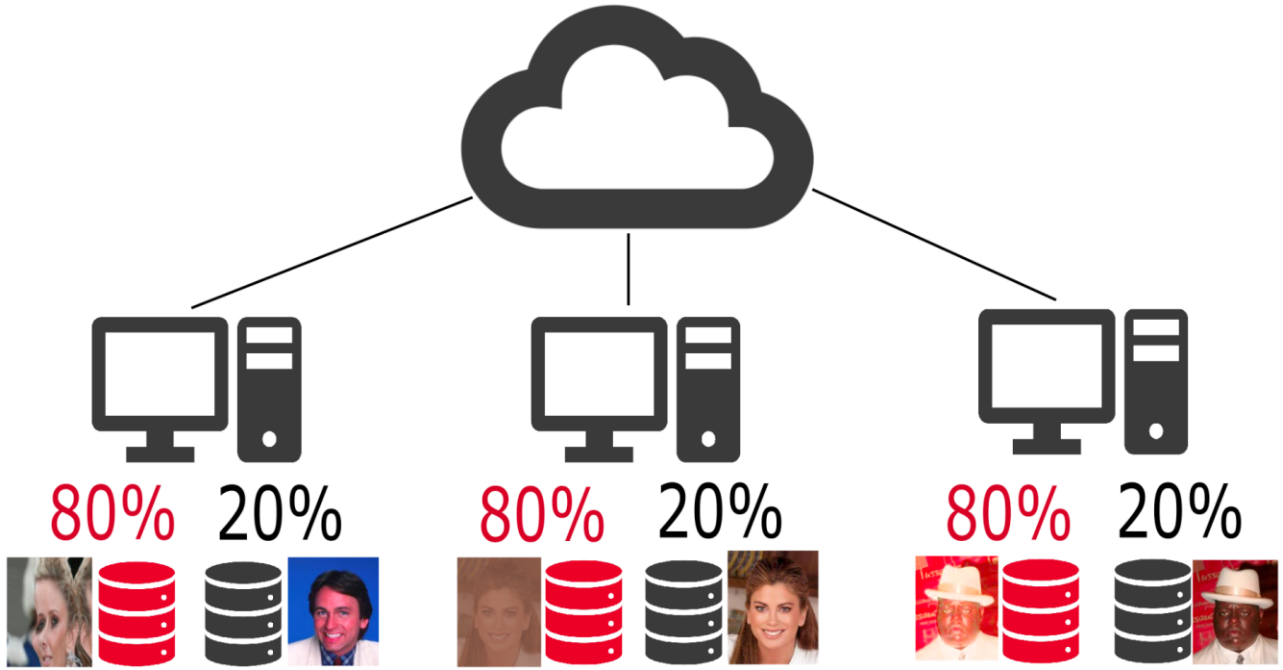
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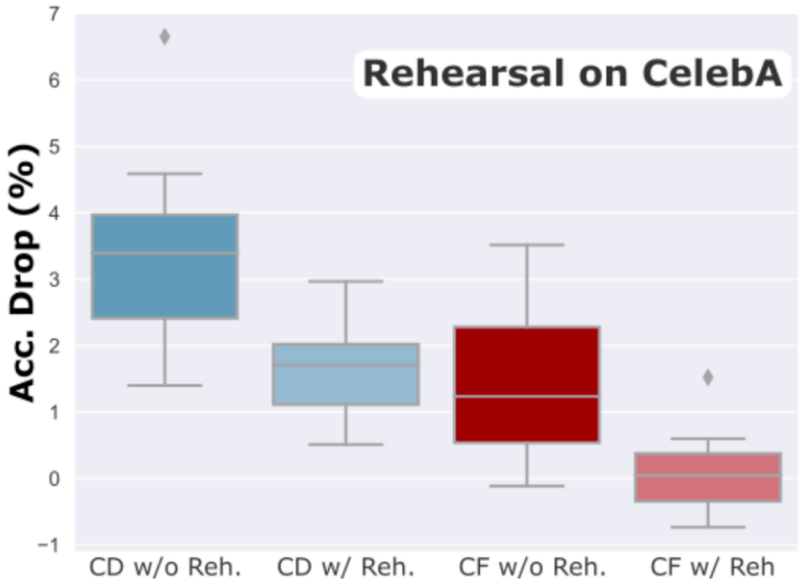
ST-Analysis: Results



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ST-Analysis: Rehearsal in FL





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ST-Analysis: Setup

	CelebA	PESO
Model	*ViT*[18]	*U-Net*[19]
Model Implementation	*ViT-FL*[20]	*BottleGAN*[21]
Federated Algorithm	*FedAvg*[22]	*FedAvg*[22]
Local Epochs	1	1
Global Epochs (CD)	200	500
Global Epochs after shift (CF)	100	40
Total Clients	227	10
Clients per Global Epoch	10	4
Seeds	[1,2,3,4,5,6,7,8,9,10]	[1,2,3]
Loss	Cross Entropy	Cross Entropy
Optimizer	SGD	Adam
Optimizer param.	-	$\beta_1 = 0.9$, $\beta_2 = 0.999$
Learning rate	0.02	0.001
Step Size	30	100
Batch Size	32	48
Image/Patch Size	178×218	114×114
GPU	NVIDIA GeForce RTX 4090	NVIDIA A40
CUDA SDK Version	12.2	11.6
PyTorch Version	2.0.0	1.13.0

Algorithm 1 Joint Analysis of CD and CF

Input: *seeds*

Require: $shifts_{cd} = [0, 1]$
  and $shifts_{cf} = [0, max_strength]$
  for *seed* in *seeds* **do**
    for $shift_{cd}$ in $shifts_{cd}$ **do**
      for $shift_{cf}$ in $shifts_{cf}$ **do**
        $model \leftarrow init(seed)$
        $perf_{cdcf} \leftarrow 0$
        for *epoch* in *cd_epochs* **do**
          $model \leftarrow train(model, shift_{cd}, seed, shift_strength = 1.0)$
        end for
        if $shift_{cd} == 0$ **and** $shift_{cf} == 0$ **then**
          $perf_{id} \leftarrow metric(test(model, data_{id}))$
        end if
        for *epoch* in *cf_epochs* **do**
          $model \leftarrow train(model, shift_ratio = 1.0, seed, shift_{strength} = shift_{cf})$
        end for
        $perf_{cdcf} \leftarrow metric(test(model, data_{id}))$
        $\blacksquare perf_{cdcf}(shift_{cd}, shift_{cf}) \leftarrow perf_{id} - perf_{cdcf}$
      end for
    end for
  end for
  return $\blacksquare perf_{cdcf}$

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DynBC: Method Details

Algorithm 1 Application of DynBC to the dynamic aggregation process to avoid Client Drift and Catastrophic Forgetting

$th_\delta \leftarrow 2.0$ {Threshold for maximum factor for exceeding the distance δ between two models (default 2.0)}
$\delta_{max} \leftarrow 0$ {Current maximum distance δ between aggregated models}
$model_{server} \leftarrow None$
for $epoch_{global}$ in $epochs_{global}$ **do**
  $newmodel_{server} \leftarrow model_{server}$
  {Check and filter Federated Learning updates from the clients for spatial continuity}
  for $model_{client}$ in $models_{clients}$ **do**
    $\delta \leftarrow DynBC(newmodel_{server}, model_{client})$ {Measure distance between client- and global model using DynBC}
    if $\delta > \delta_{max}$ **then**
      if $\delta <= th_\delta * \delta_{max}$ **then**
        $newmodel_{server} \leftarrow Aggregate(newmodel_{server}, model_{client})$
        $\delta_{max} \leftarrow \delta$
      end if
      $\delta_{max} \leftarrow \delta$
    else
      $newmodel_{server} \leftarrow Aggregate(newmodel_{server}, model_{client})$
    end if
  end for
  {Check and filter Continual Learning updates for temporal continuity}
  if $DynBC(model_{server}, newmodel_{server}) \leq th_\delta * \delta_{max}$ **then**
    $model_{server} \leftarrow newmodel_{server}$
  end if
end for

DynBC: Results

Dataset	BCSS		Semicol	
Shift Type	Gaussian Blur	Brightness	Gaussian Blur	Brightness
Baseline	0.518 ± 0.13	0.158 ± 0.139	0.649 ± 0.04	0.034 ± 0.028
DynBC (Ours)	$\mathbf{0.849 \pm 0.007}$	$\mathbf{0.715 \pm 0.033}$	$\mathbf{0.73 \pm 0.05}$	$\mathbf{0.227 \pm 0.326}$
Rehearsal	0.09 ± 0.11	0.124 ± 0.203	0.031 ± 0.029	0.209 ± 0.12

Table 1. Evaluation in Client Drift scenario with mean and standard deviation of dice scores

Dataset	BCSS		Semicol	
Shift Type	Gaussian Blur	Brightness	Gaussian Blur	Brightness
Baseline	0.469 ± 0.222	0.425 ± 0.245	0.194 ± 0.138	0.634 ± 0.018
DynBC (Ours)	0.709 ± 0.049	0.628 ± 0.308	0.455 ± 0.126	0.625 ± 0.031
Rehearsal	$\mathbf{0.82 \pm 0.007}$	$\mathbf{0.822 \pm 0.002}$	$\mathbf{0.741 \pm 0.042}$	$\mathbf{0.738 \pm 0.039}$

Table 2. Evaluation in Catastrophic Forgetting scenario with mean and standard deviation of dice scores

DynBC: Experiment Details

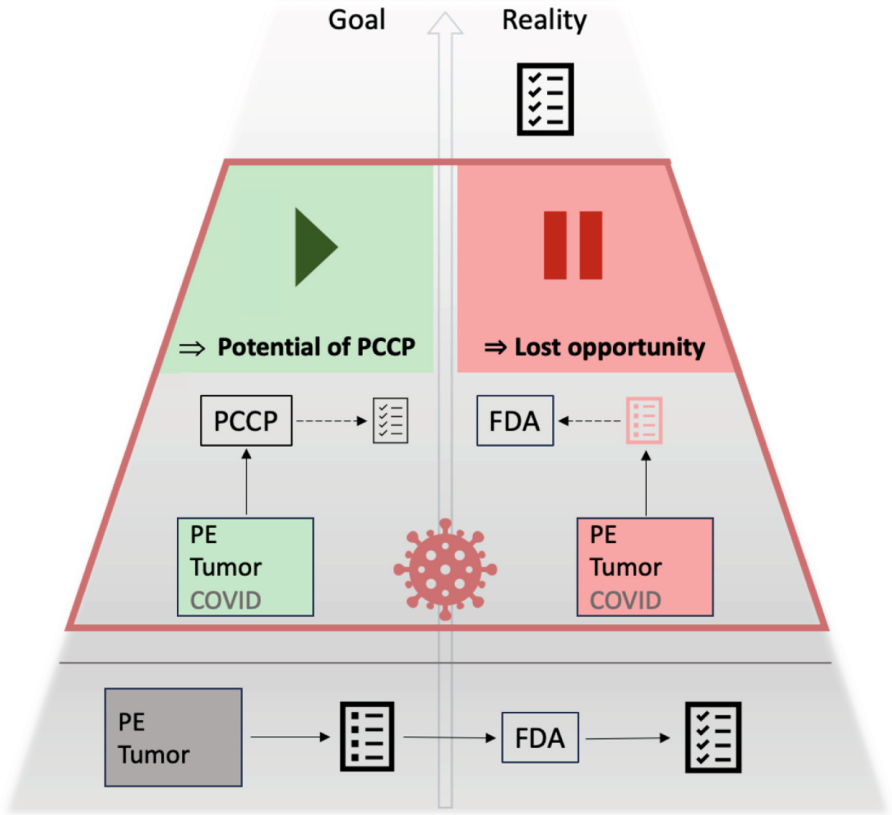
Reference Set:
- 1125 patches from Camelyon 17
- 256 pixel
- 0.25 microns per pixel
- Augmentations
- Torchvision Augmentations:
  - Gaussian Blur (only when we don't evaluate on blur!)
  - Gaussian Noise
- Albumentations
  - Motion Blur (only when we don't evaluate on blur!)

BCSS Evaluation:
- 60% train, 30% test, 20% validation
- 256 pixel
- 0.25 microns per pixel

SemiCOL Evaluation:
- 70% train, 20% test, 10% validation
- 256 pixel
- 0.25 microns per pixel

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PCCP: Intended Use Gap

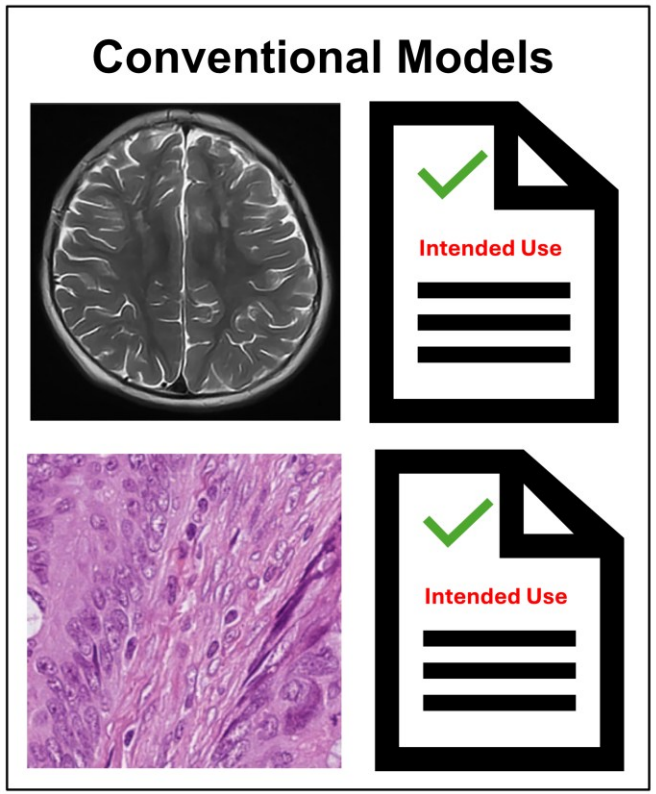


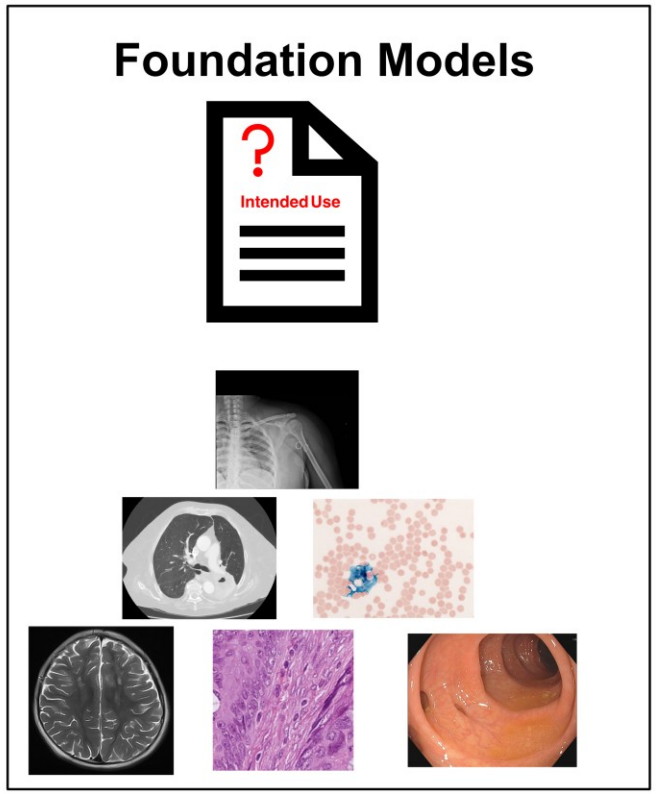
Gap:
Continual Learning could rapidly adapt AI medical devices to new tasks like pandemic diseases, but PCCP's strict "intended use" rule blocks such updates, delaying critical deployments.

Solution:
Expanding PCCP's "intended use" to allow adding new diseases with shared manifestations under existing modalities and task types would enable faster, safer SaMD updates during crises like pandemics.

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PCCP: Foundation Model Gap





Gap:
PCCP's strict intended use requirement makes FDA approval impractical for versatile medical Foundation Models, limiting their potential to advance AI-assisted SaMDs.

Solution:
Broadening SaMD intended use to cover groups of tasks and modalities for Foundation Models would enable safe, flexible deployment while preserving rigorous performance standards.

PCCP: Situation worldwide

EU AI Act in combination with Medical Device Regulation (MDR):
- High risk AI should allow controlled changes for BIAS-avoidance
- Should be part of Quality Management System (QMS)
- Still far beyond what PCCP allows in the United States or adopting countries

Health Canada, MHRA (UK) and MFDS (Republic of Korea):
- Adopting changes similar to PCCP

PMDA (Japan):
- Checking change protocol provided manufacturer

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Nationwide FL: Setup

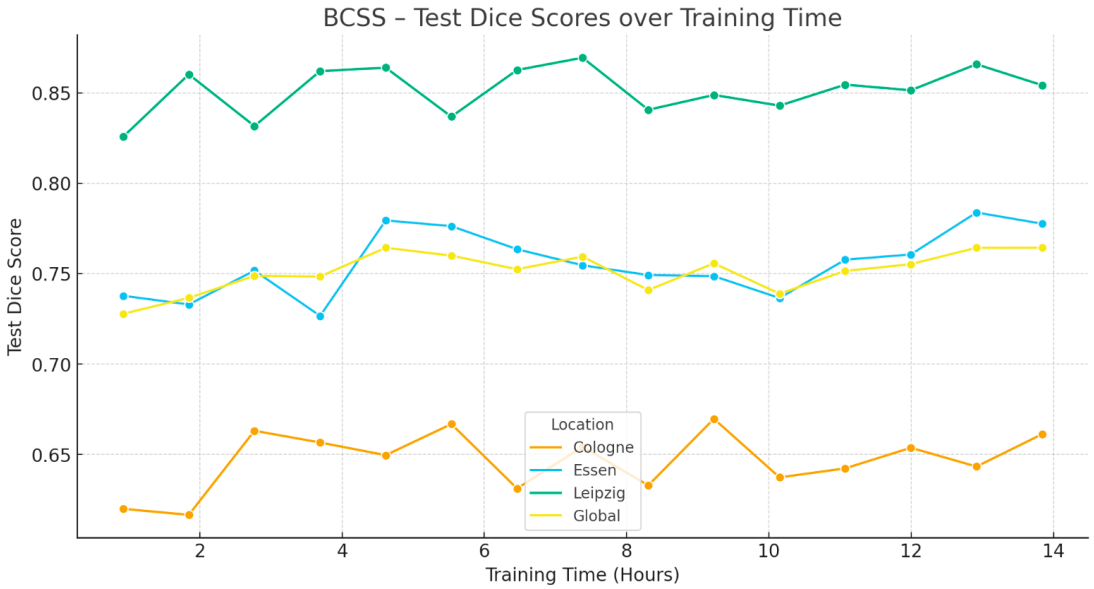
	Server	Reverse Proxy (Virtual Machine)	Clients
GPU	NVIDIA GeForce RTX 4090 with 24 GB VRAM	No GPU	NVIDIA GeForce RTX 4090 with 24 GB VRAM
CPU	Intel Core i7-13700K (8 Cores, 3,4 GHz)	8 vCPUs	AMD Ryzen 9 5950X (16 Cores, 3.4 GHz)
RAM	64 GB DDR4 RAM	32 GB RAM	128 GB DDR4 RAM
Storage	2 TB Kingston Fury Renegade 2280 NVMe SSD	100 GB SSD	2 TB Samsung 870 QVO SSD
OS	Ubuntu 22.04 LTS	Ubuntu 22.04 LTS	Ubuntu 22.04 LTS

Table 1. Hardware deployed at the server and reverse proxy in Darmstadt and the clients in Cologne, Essen and Leipzig.

Client	BCSS Train	BCSS Validation	BCSS Test	SemiCOL Train	SemiCOL Validation	SemiCOL Test
Cologne	34	7	7	3 (UKK)	1 (UKK)	1 (UKK)
Essen	34	7	7	3 (UKK)	1 (UKK)	1 (UKK)
Leipzig	34	7	7	6 (LMU)	2 (LMU)	2 (LMU)
Total	102	21	21	12	4	4

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Nationwide FL: Results



(a) BCSS



(b) SemiCOL

Nationwide FL: Results

Table 1. Dice score per class for each label of the SemiCOL dataset per location and globally.

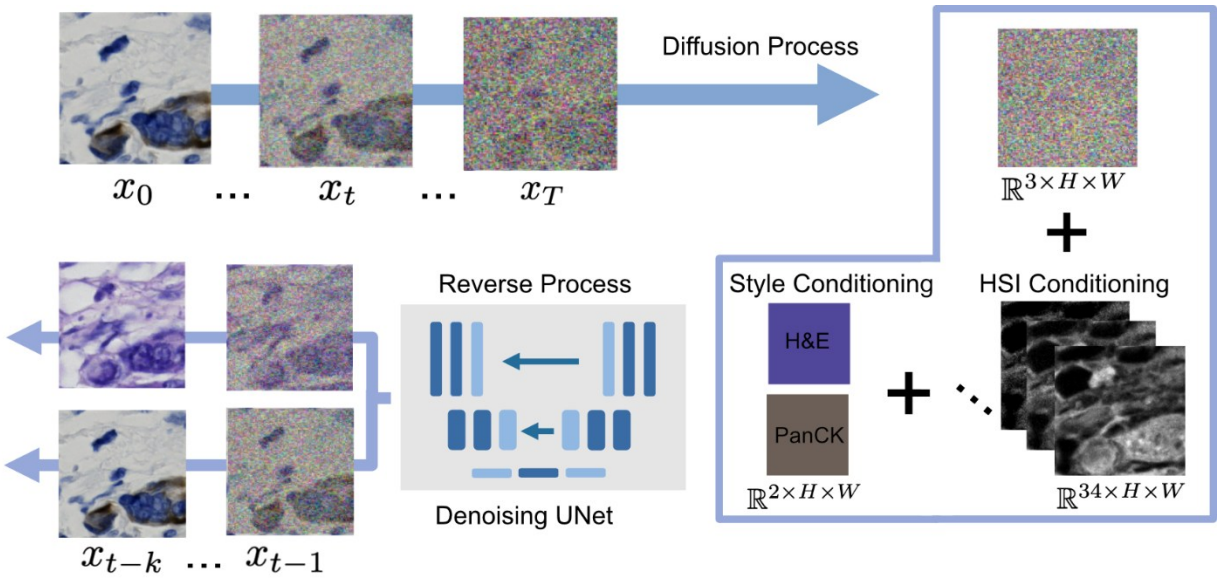
Class	0	1	2	3	4	5	6	7	8	9
Cologne	0,9515	0,984	0,5681	0,9079	0,9722	0,7981	0,9509	0,3105	0,8067	0,9702
Essen	0,8884	0,9871	0,7827	0,9364	0,901	0,973	0,8271	0	0,2273	0,9603
Leipzig	0,8446	0,9119	0,7226	0,7875	0,4434	0,799	0,9085	0	0,5714	0,9141
Global	0,8948	0,9634	0,6911	0,8773	0,7722	0,8567	0,8955	0,1035	0,5335	0,9482

Table 2. Dice score per class for each label of the BCSS dataset per location and globally.

Class	0	1	2	3
Cologne	0,8927	0,6477	0,6477	0,4086
Essen	0,9018	0,84	0,6571	0,7115
Leipzig	0,902	0,8583	0,7575	0,7322
Global	0,9113	0,7906	0,6933	0,6621

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FedVS4Hist: Results



Method	Pathologist 1	Pathologist 2	Overall
StarGAN v2	0%	6%	3%
VISTA (Ours)	**100%**	**94%**	**97%**

Method	FID (↓)	KID (↓)	LPIPS div. (↑)
StarGAN v2	220.450	0.190 ± 0.004	0.297 ± 0.104
VISTA (Ours)	**99.883**	$\mathbf{0.089 \pm 0.004}$	$\mathbf{0.379 \pm 0.099}$